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(54) **WAIT WATER COLLECTION SYSTEM**

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(60) Provisional application No. 63/559,263, filed on Feb. 29, 2024, provisional application No. 63/559,288, filed on Feb. 29, 2024.

(57) **ABSTRACT**

The wait water collection system provided herein may be used to reducing the amount of water wasted while a shower, for example, warms up. The wait water collection system may include a hot water input carrying hot water to the system. The hot water may or may not be “hot” when delivered by the hot water input and may need to “warm up.” The wait water collection system may include a thermal diverter on a water line where the hot water travels, which may adjust based on whether the temperature of the hot water meets a temperature threshold. If the water does not meet the temperature threshold, then the hot water may be delivered to an expansion vessel. If the water meets or exceeds the temperature threshold, then the hot water may be delivered to a mixer valve for providing the water to a user.

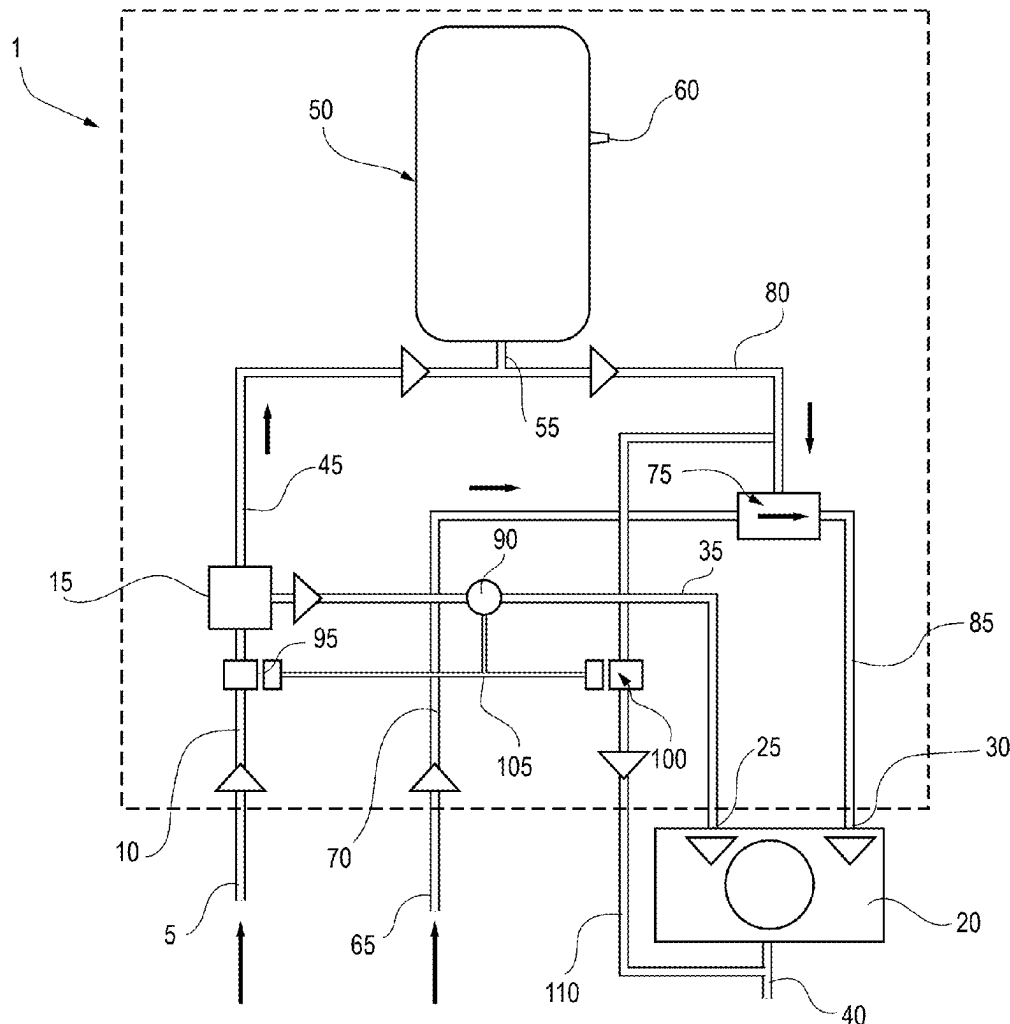


FIG. 1

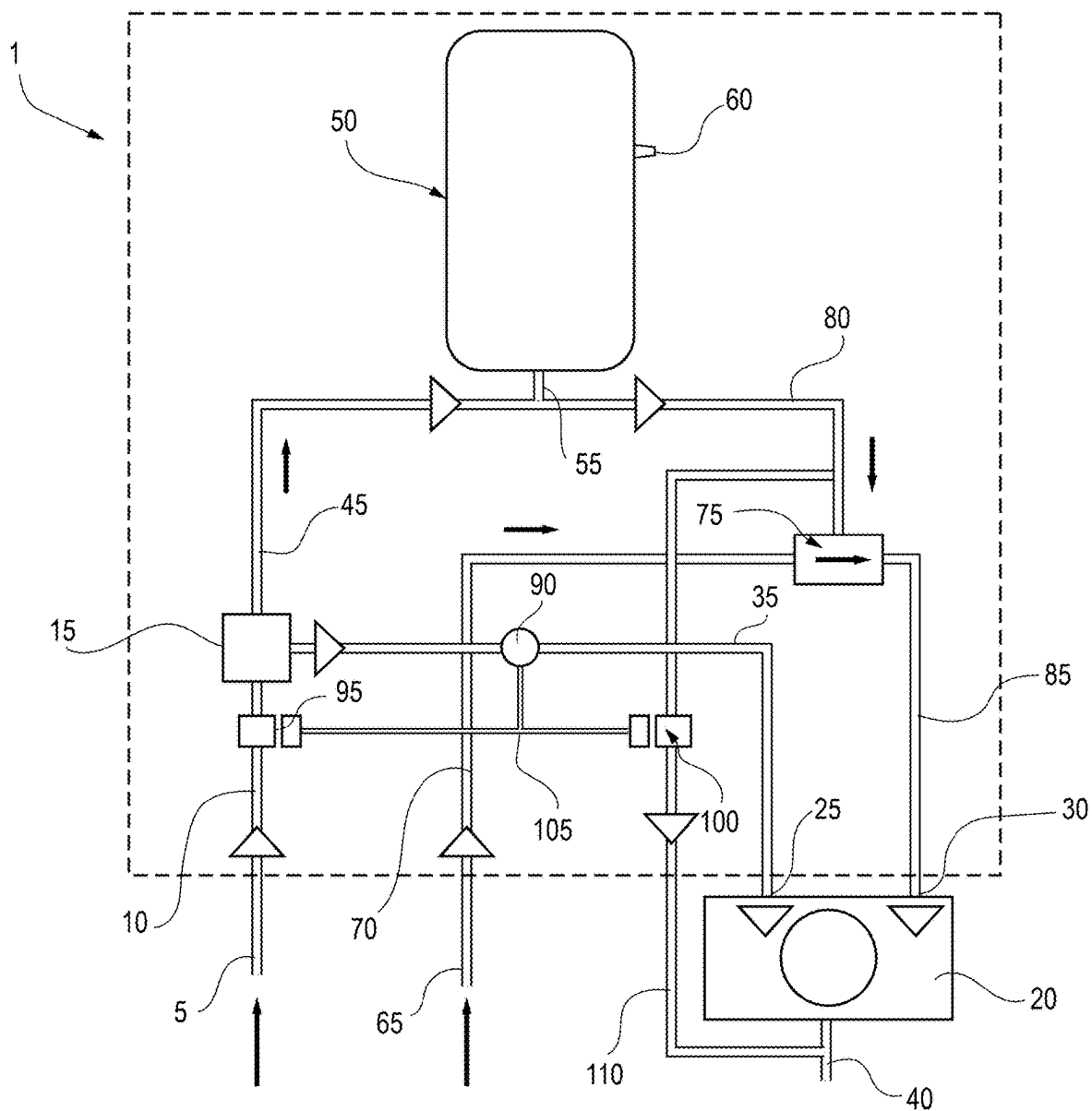
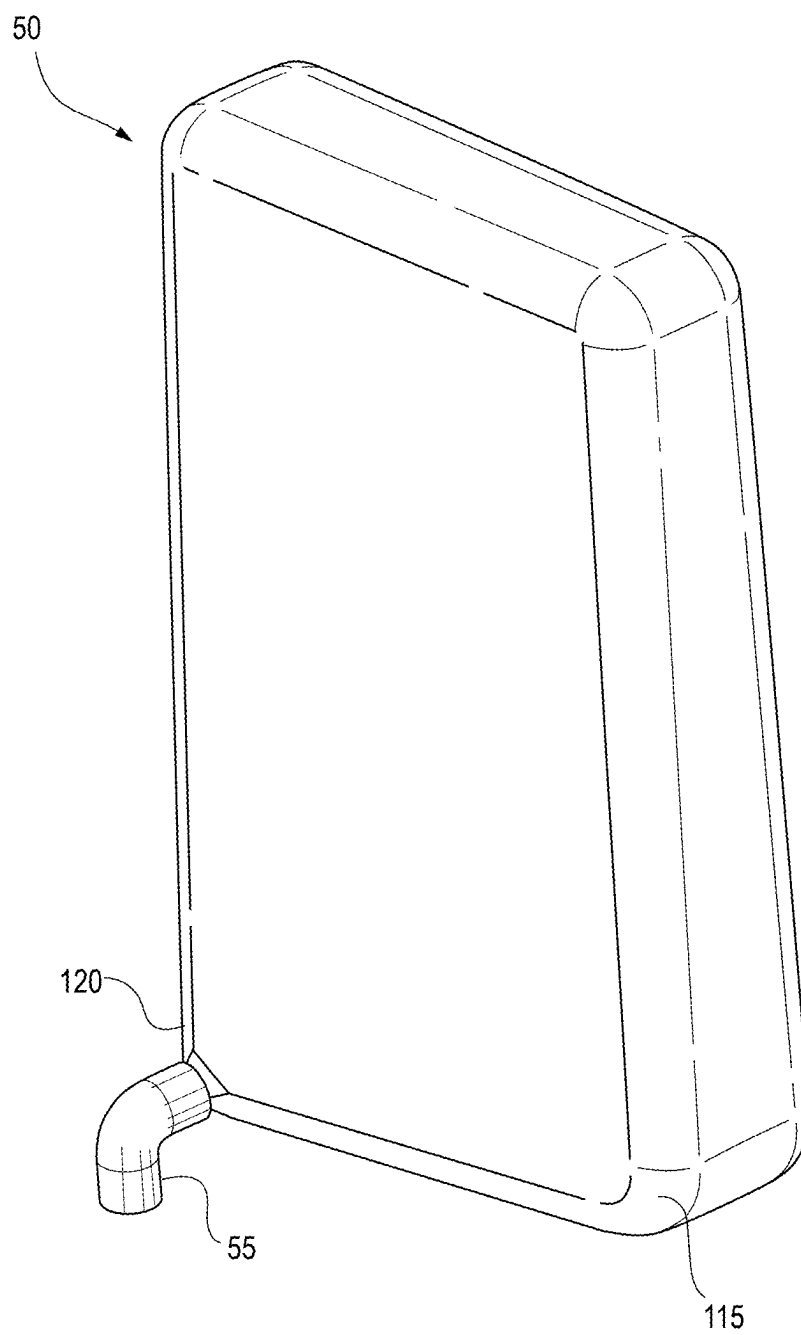


FIG. 2

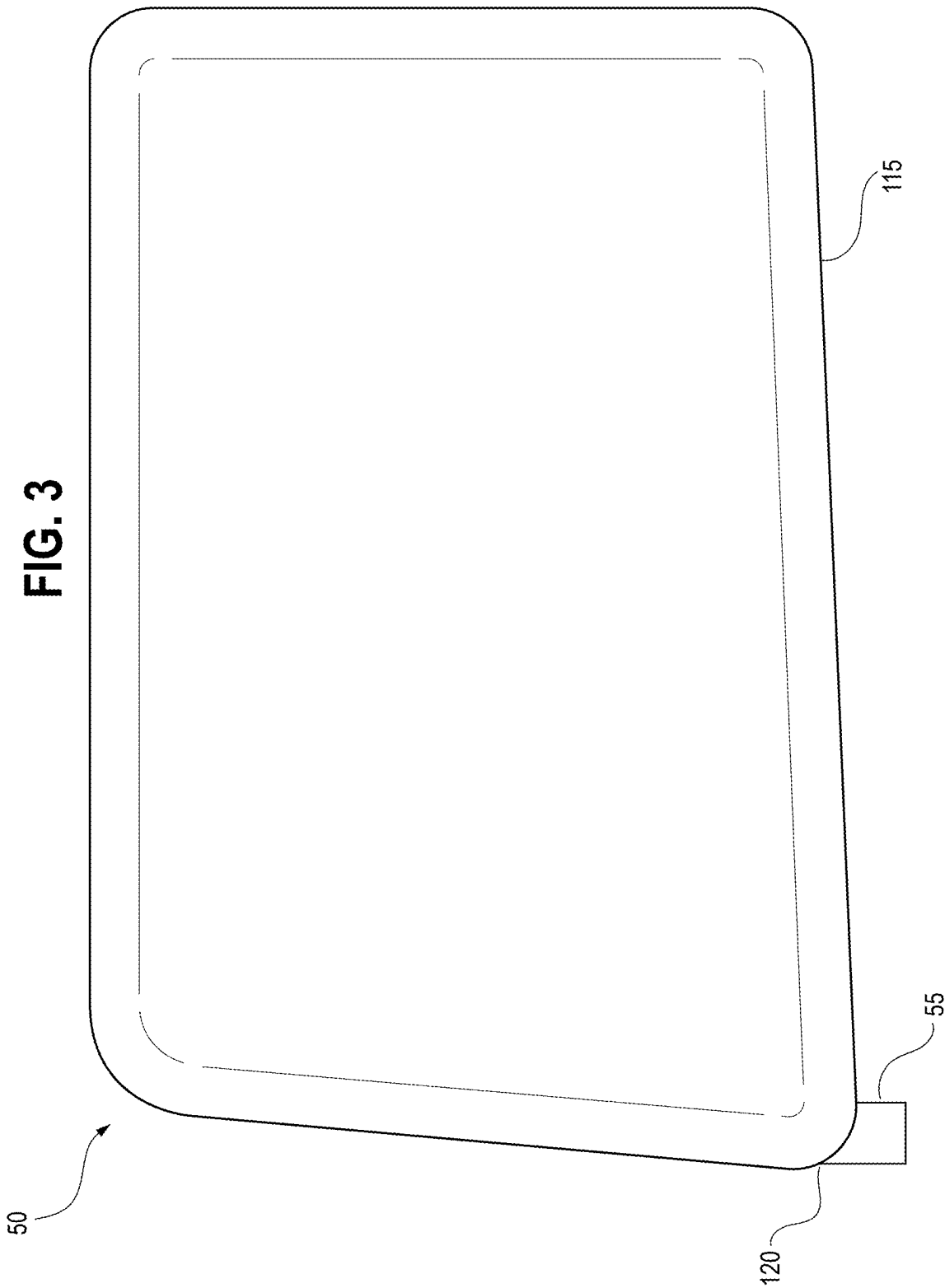


FIG. 4

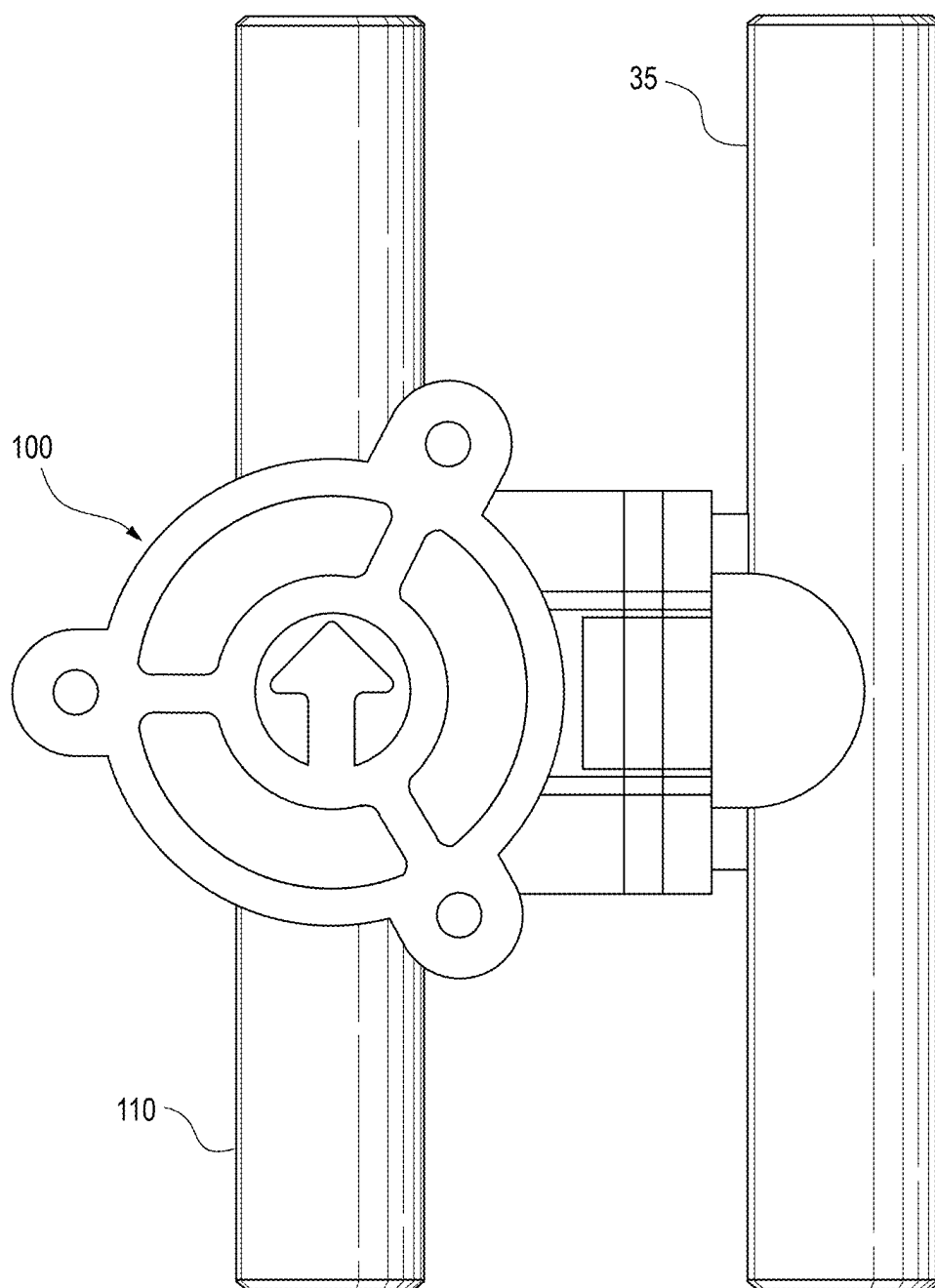


FIG. 5

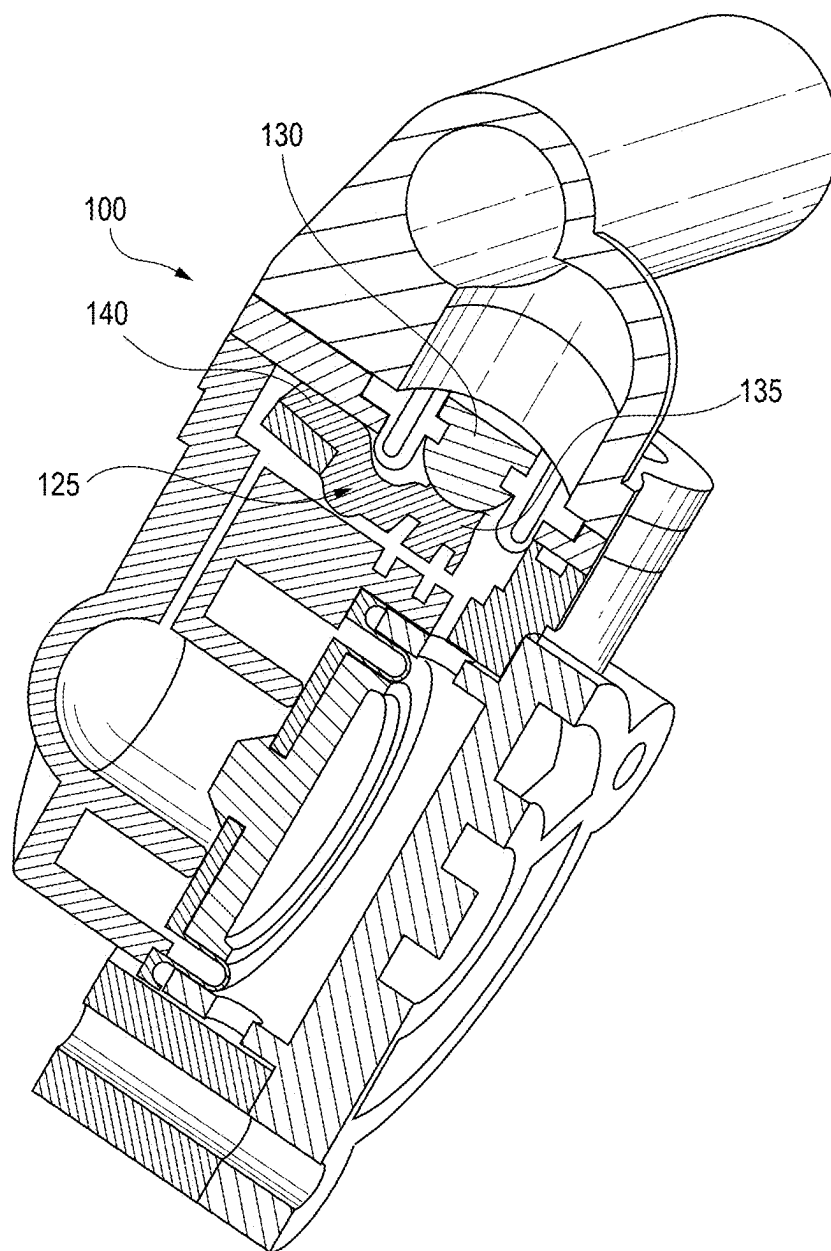
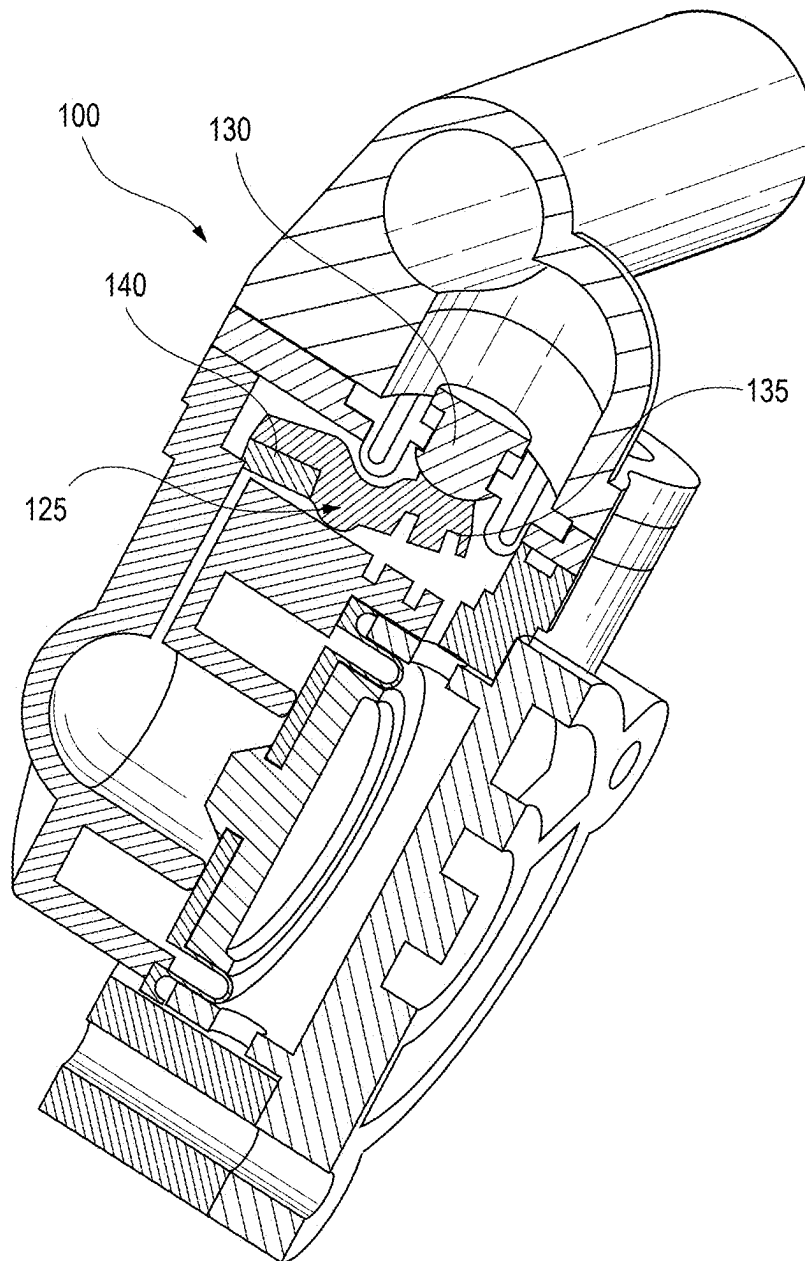
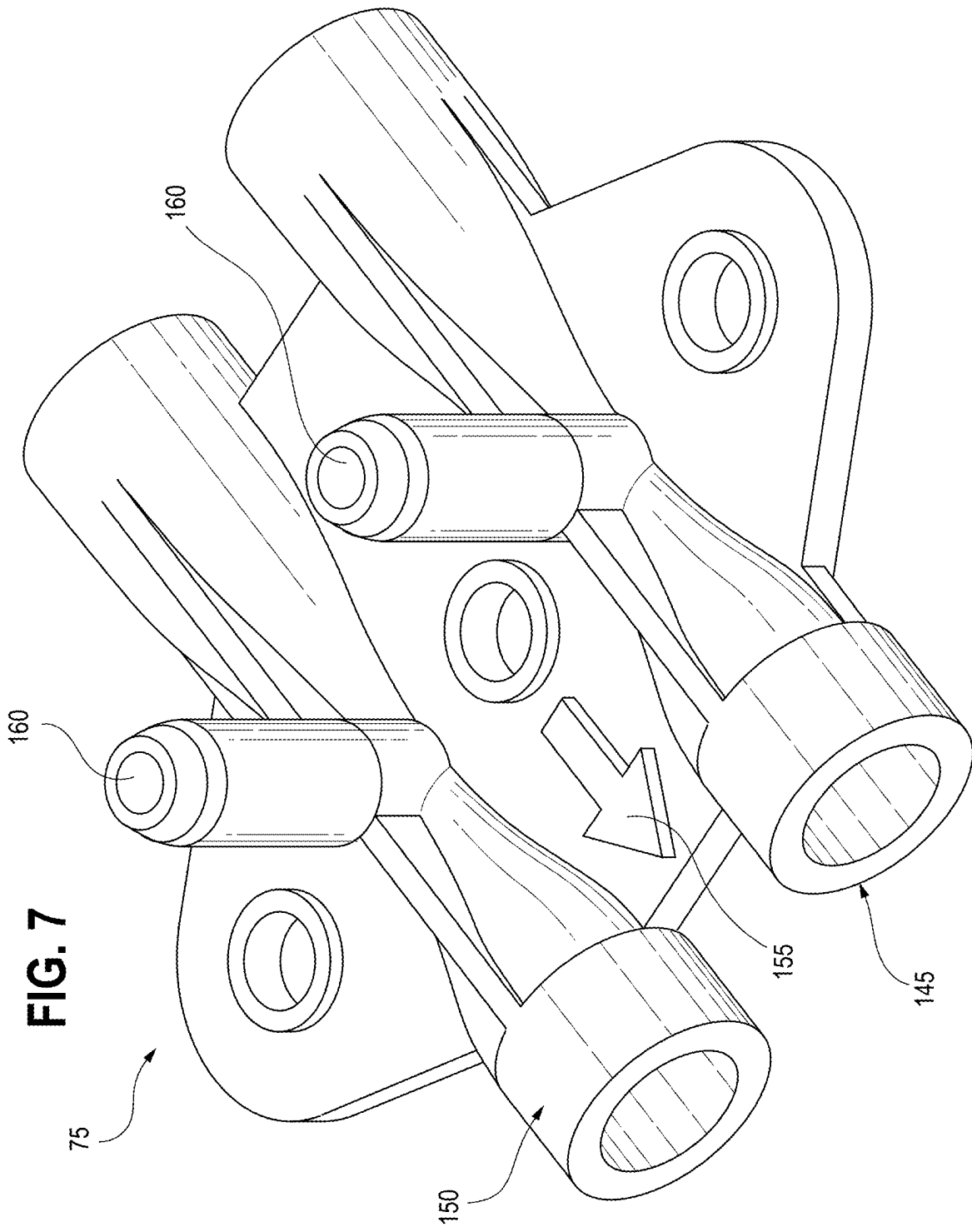


FIG. 6





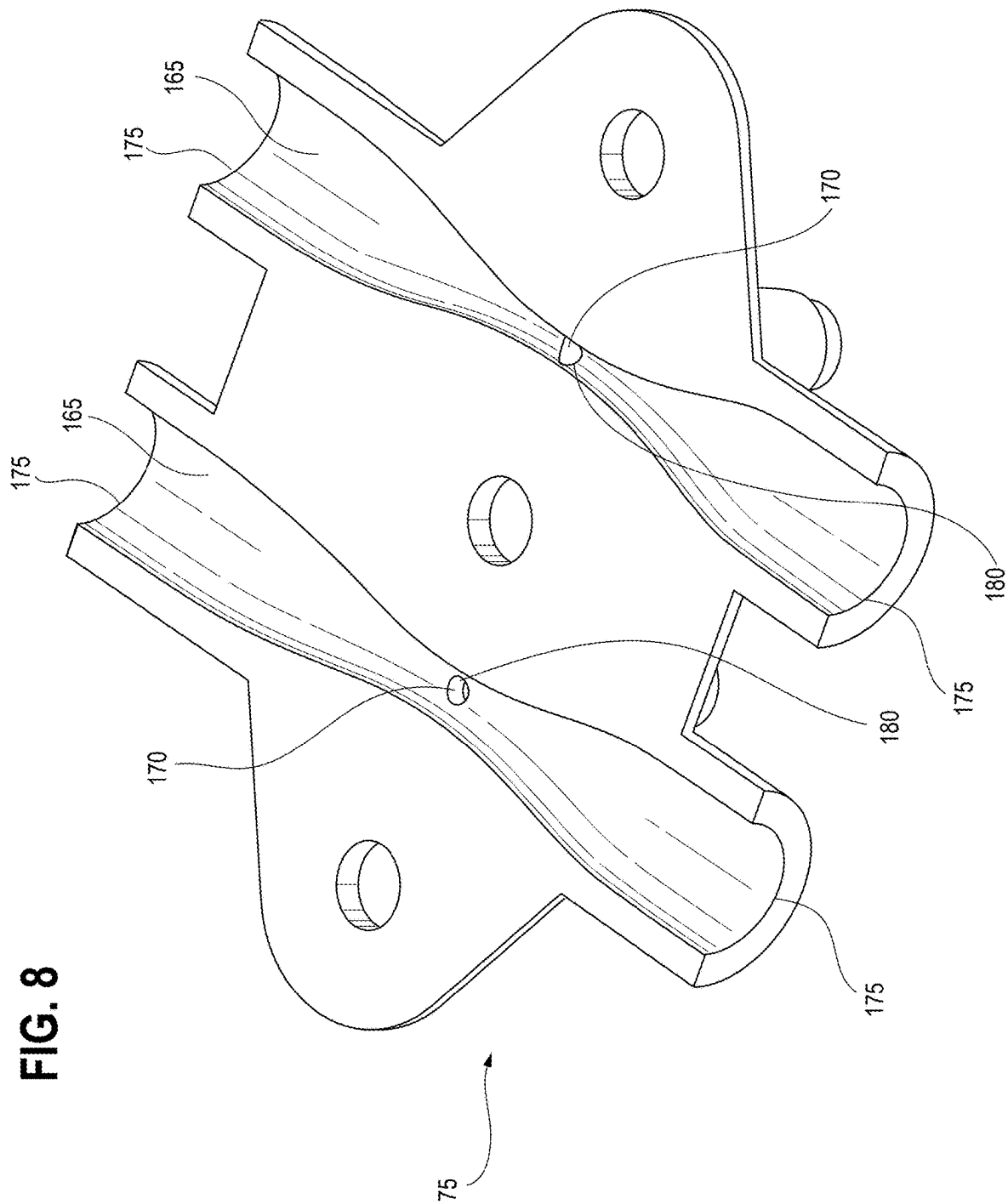


FIG. 9

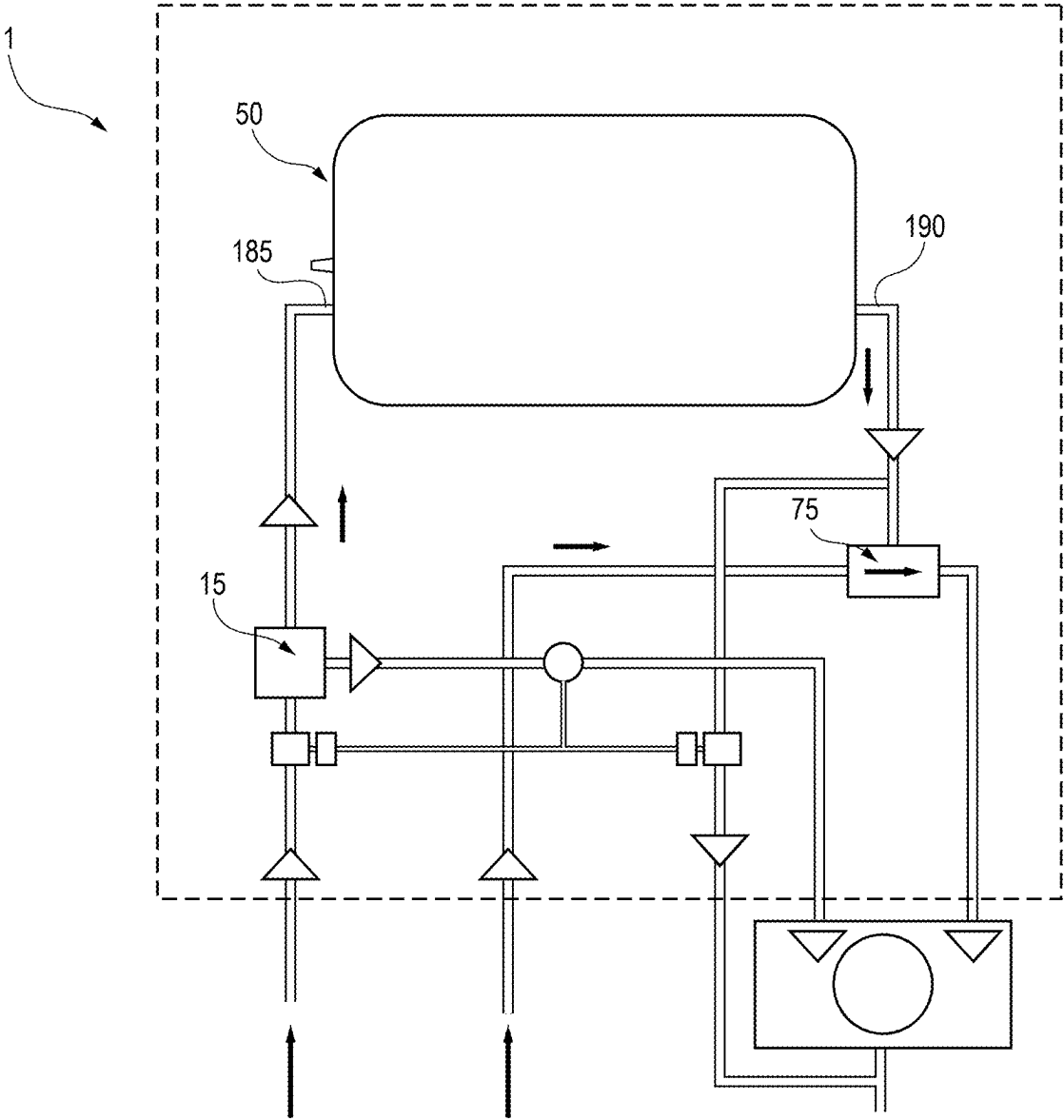


FIG. 10

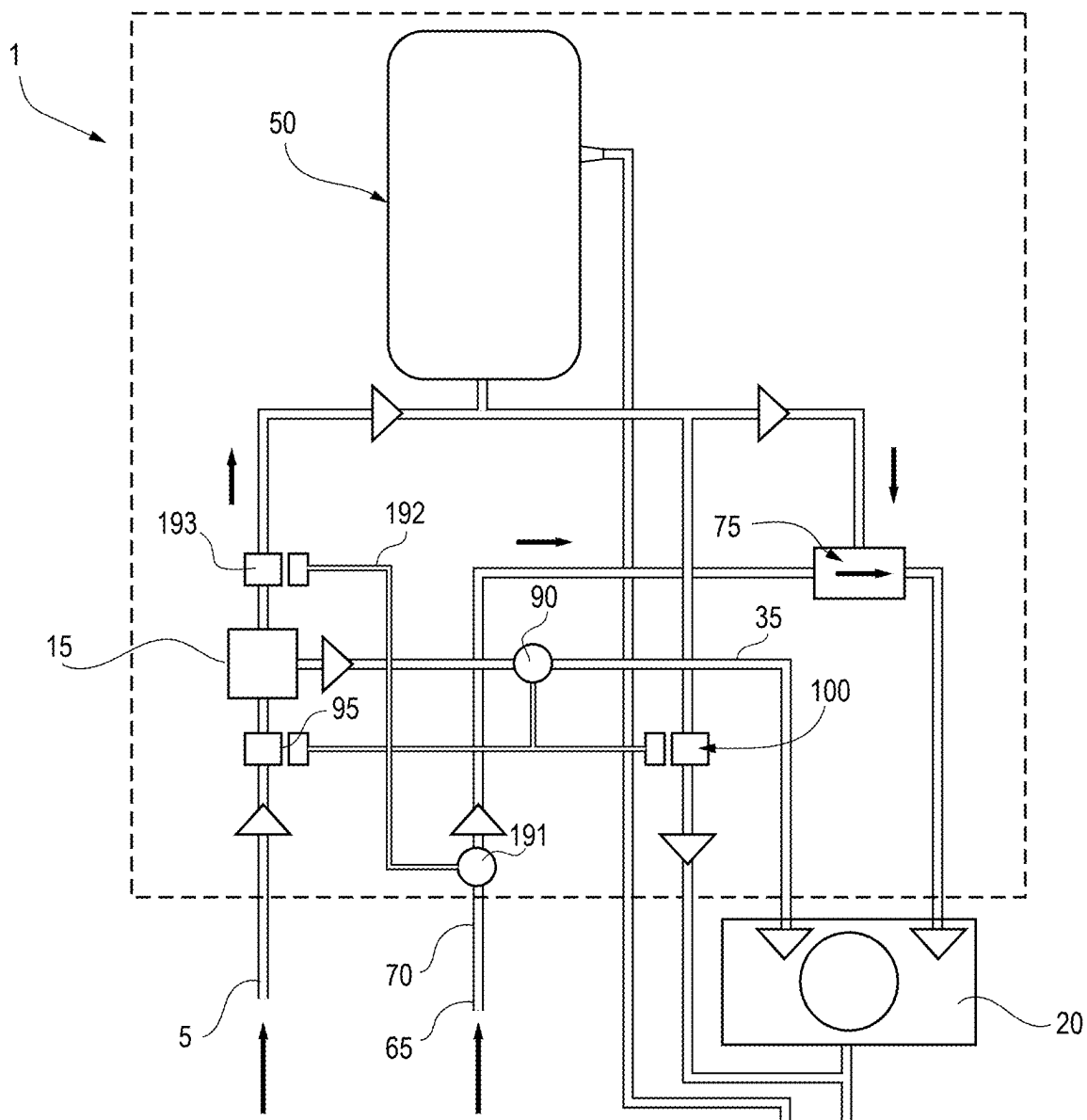


FIG. 11

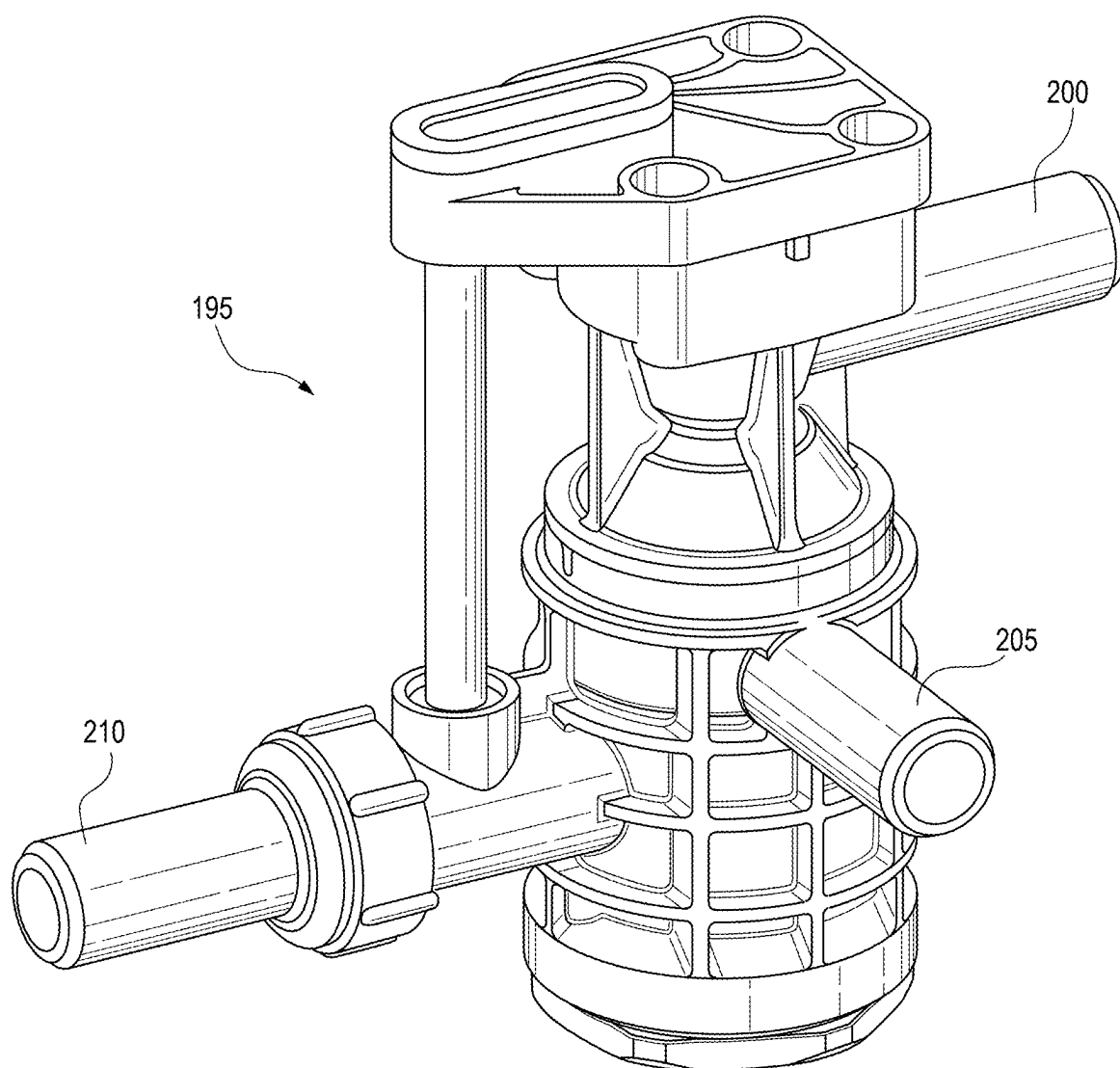
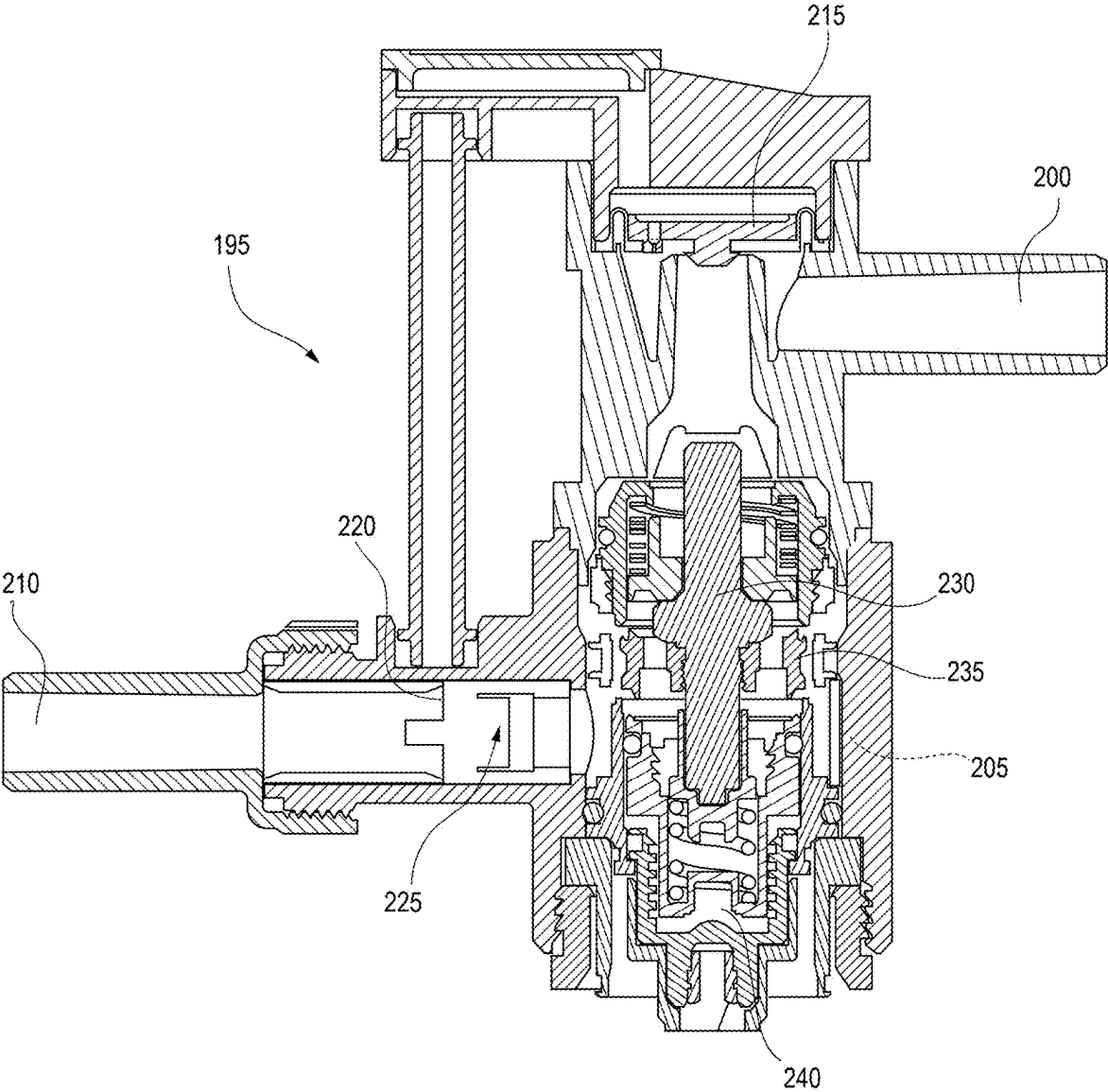


FIG. 12



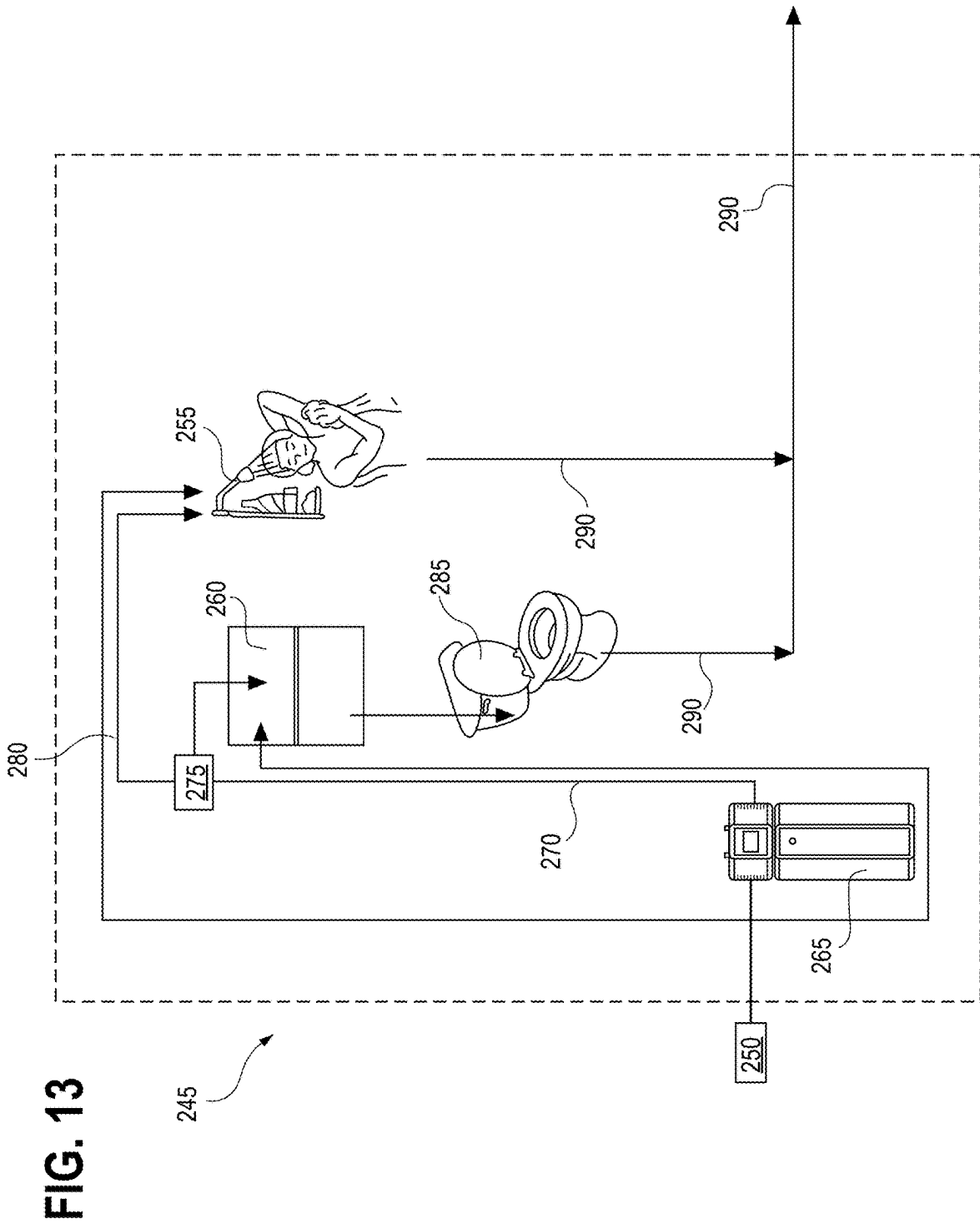


FIG. 14

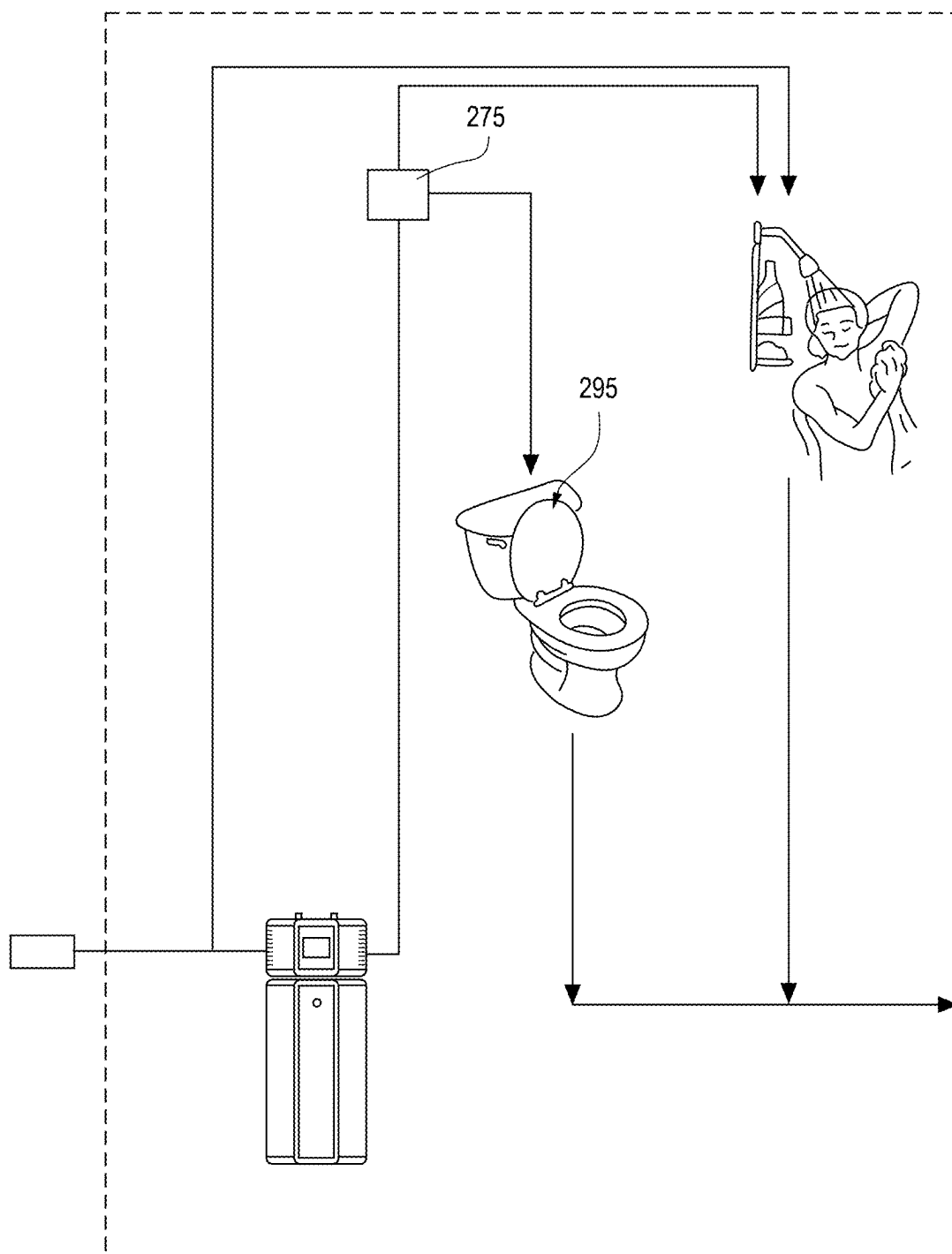


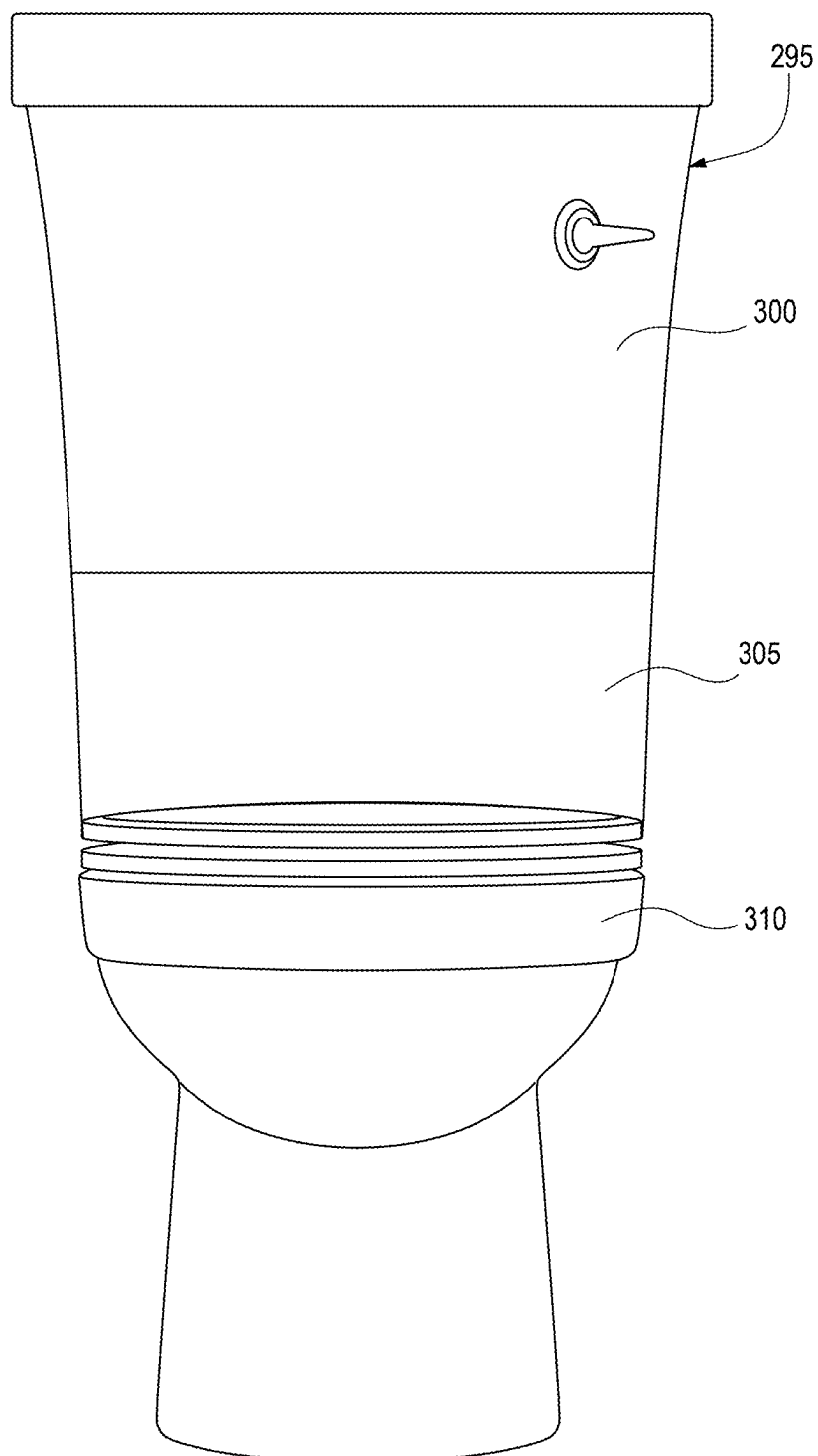
FIG. 15

FIG. 16

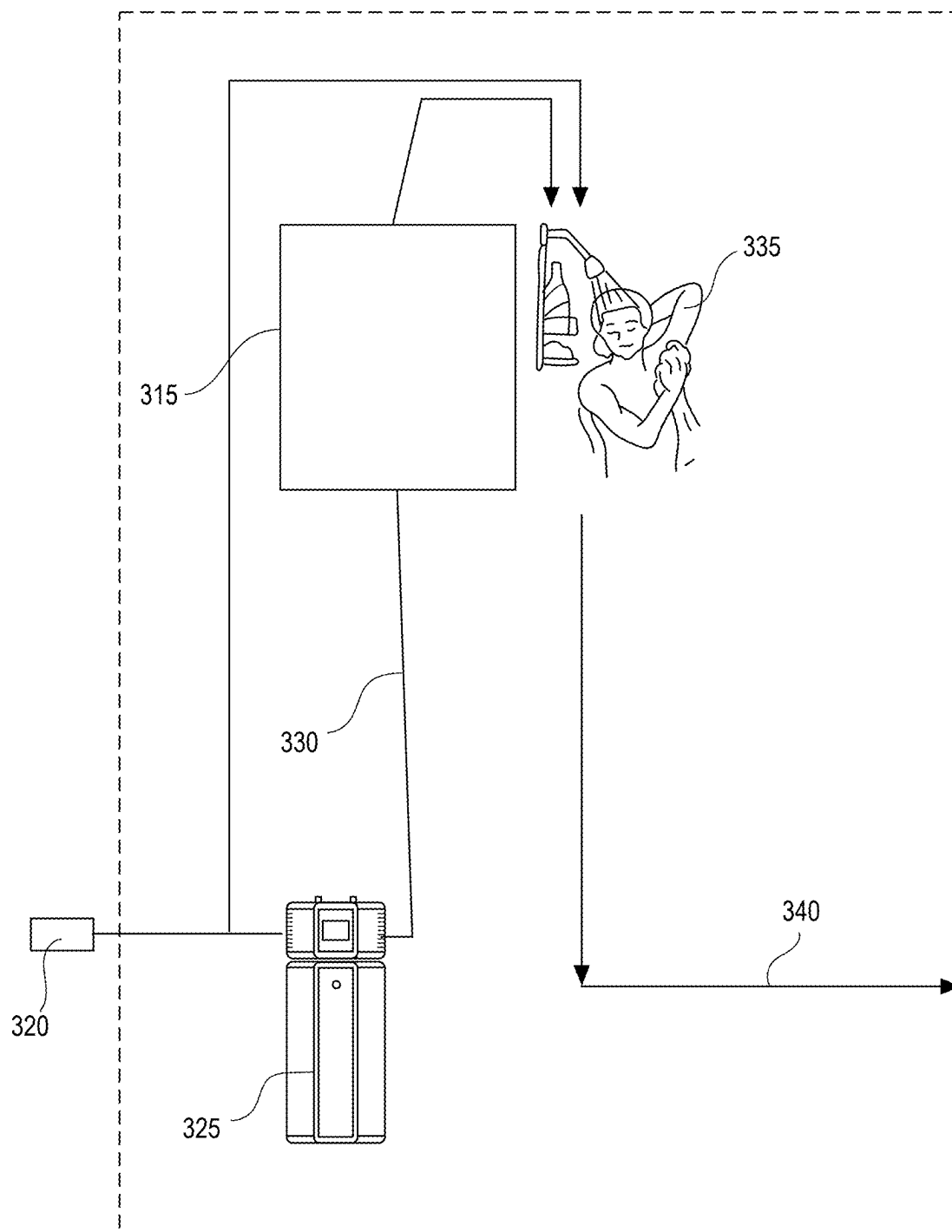


FIG. 17

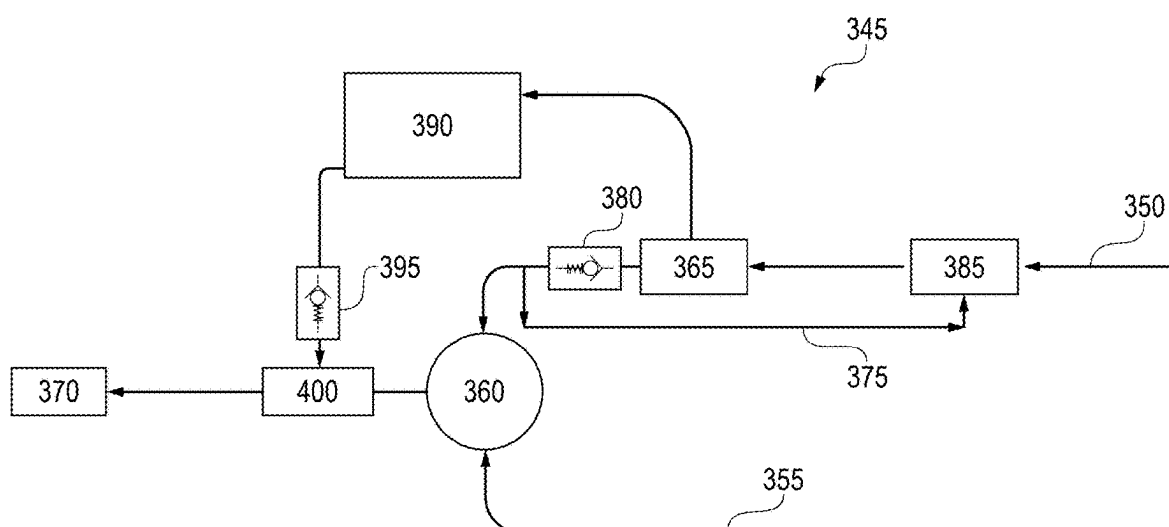


FIG. 18

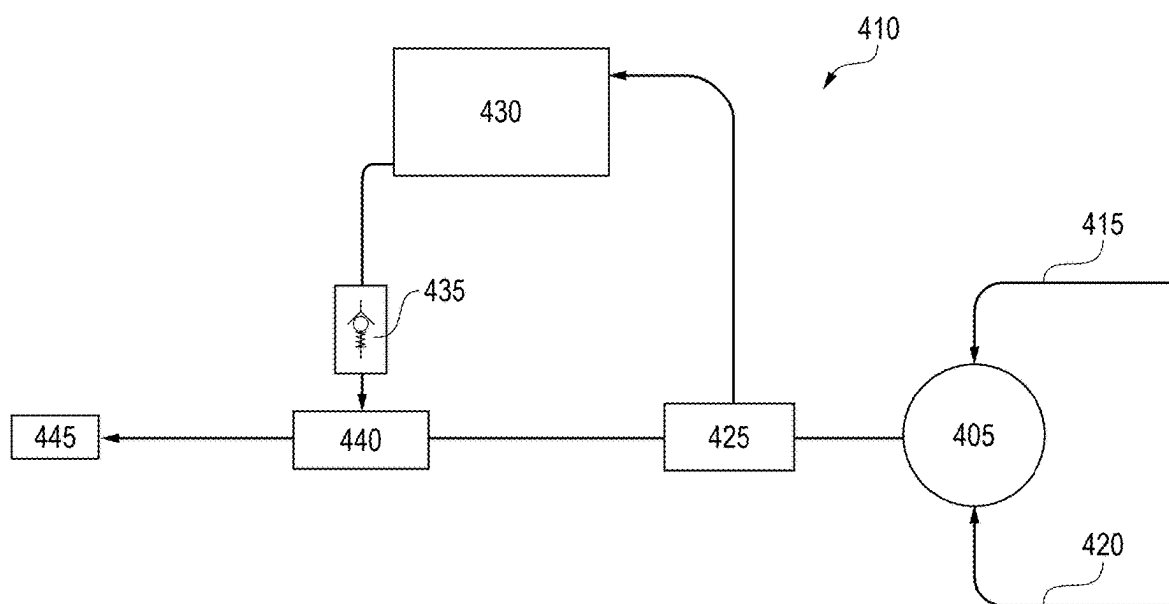


FIG. 19

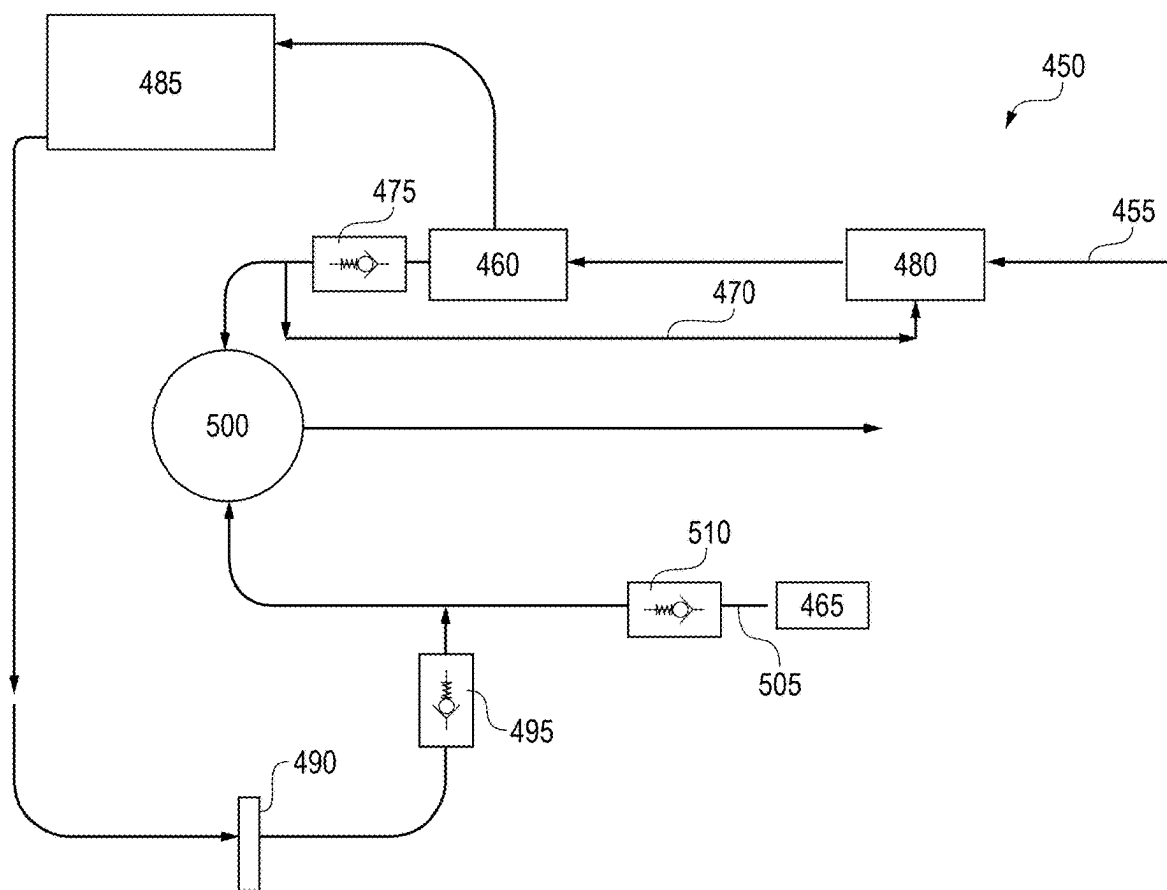


FIG. 20

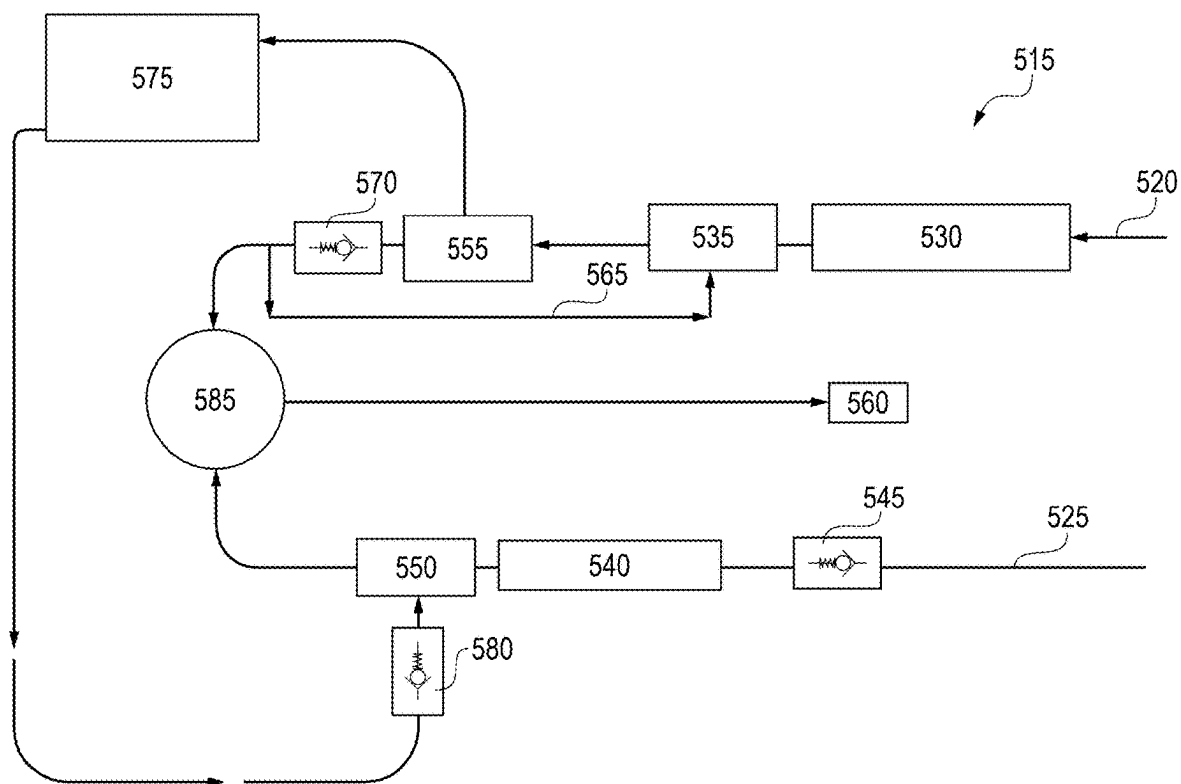


FIG. 21

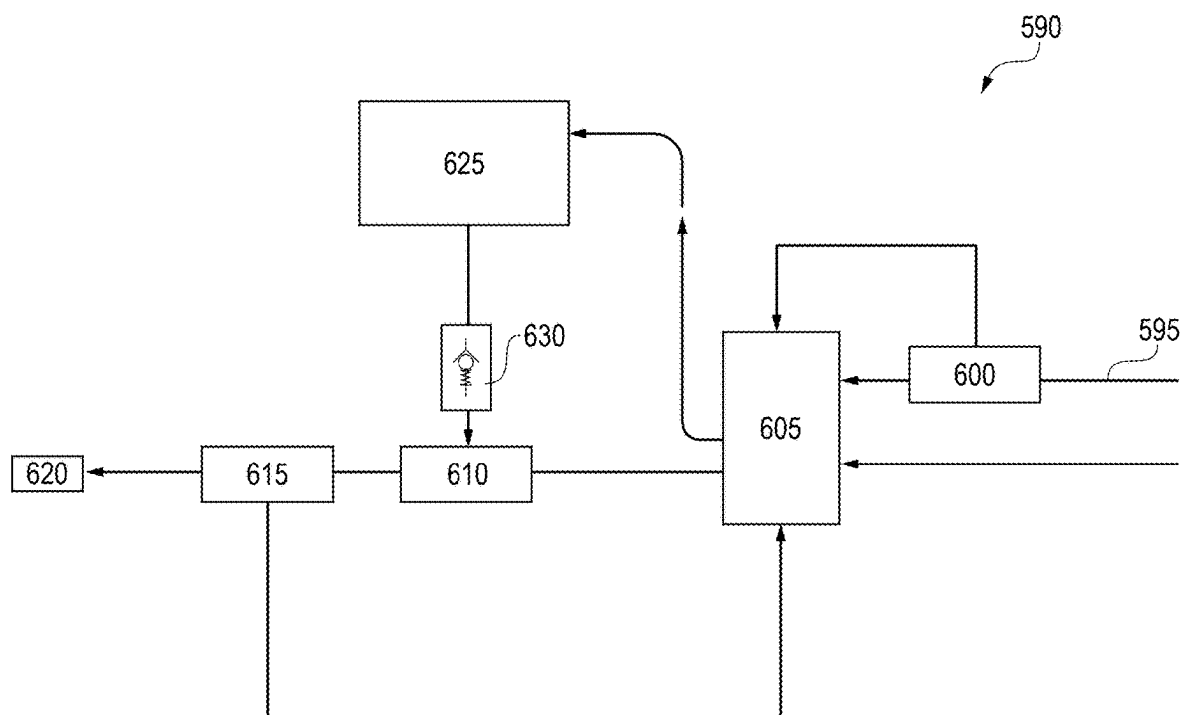


FIG. 22

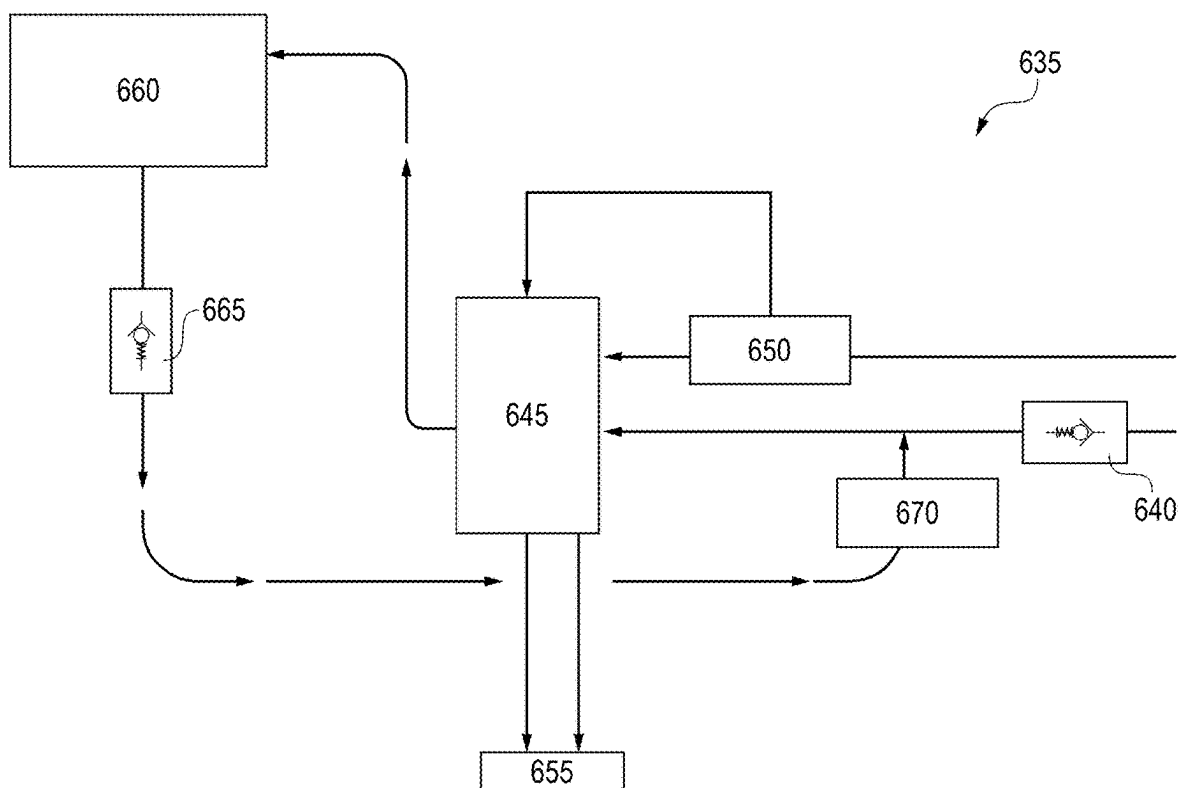


FIG. 23

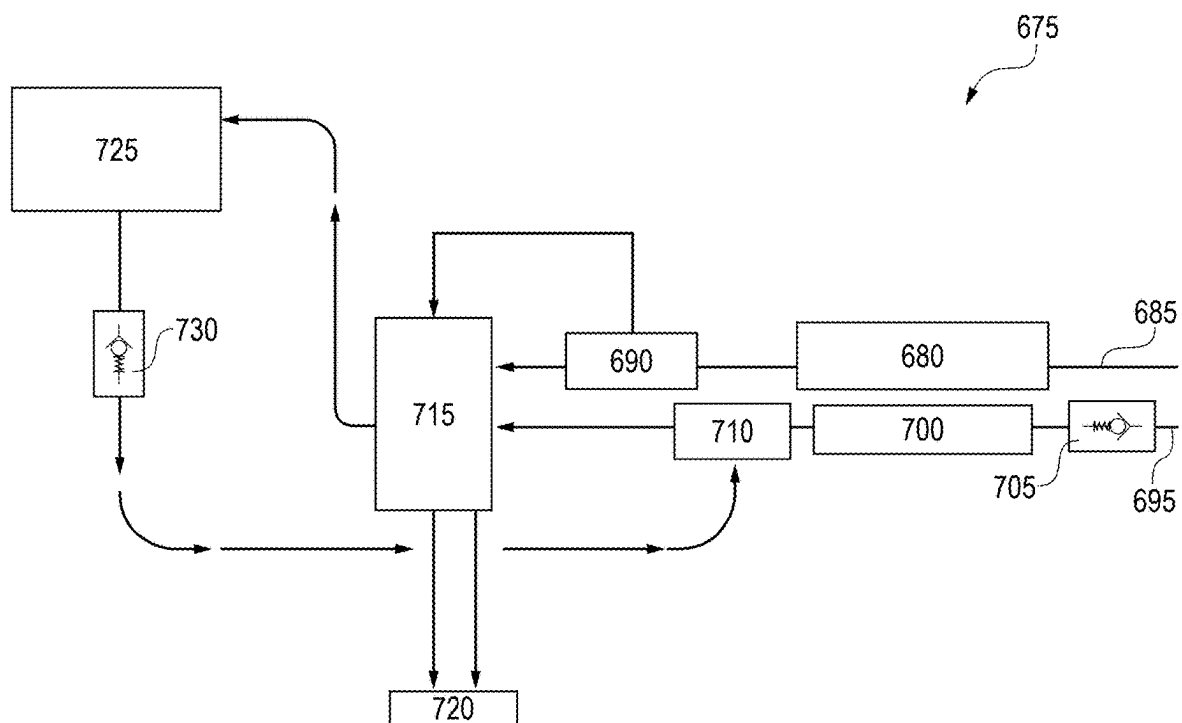


FIG. 24

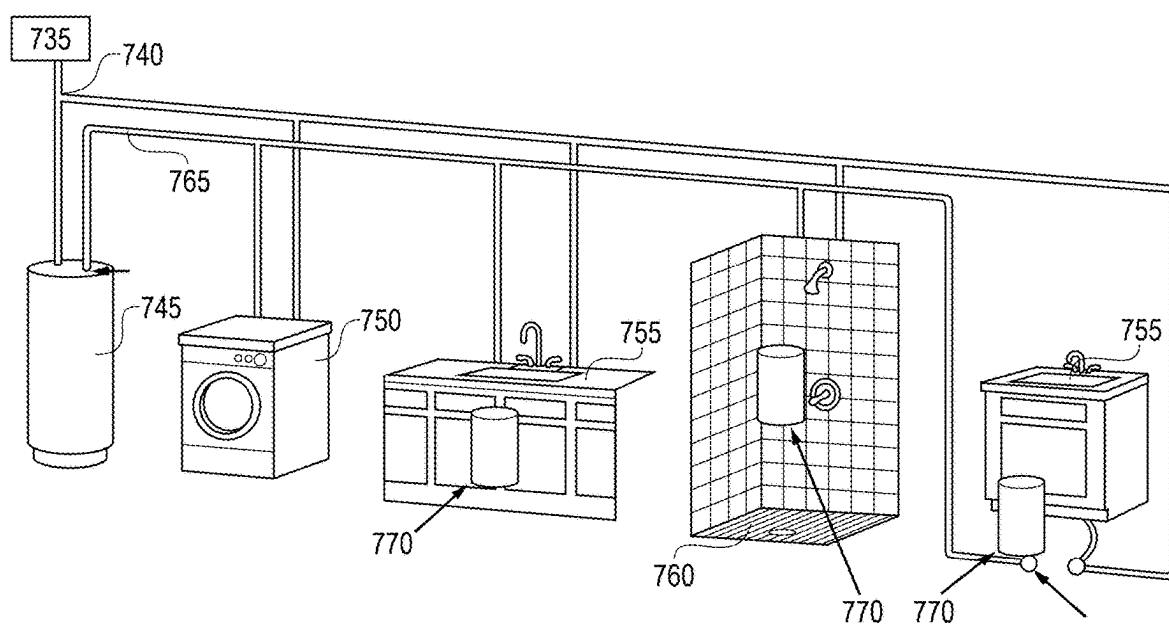


FIG. 25

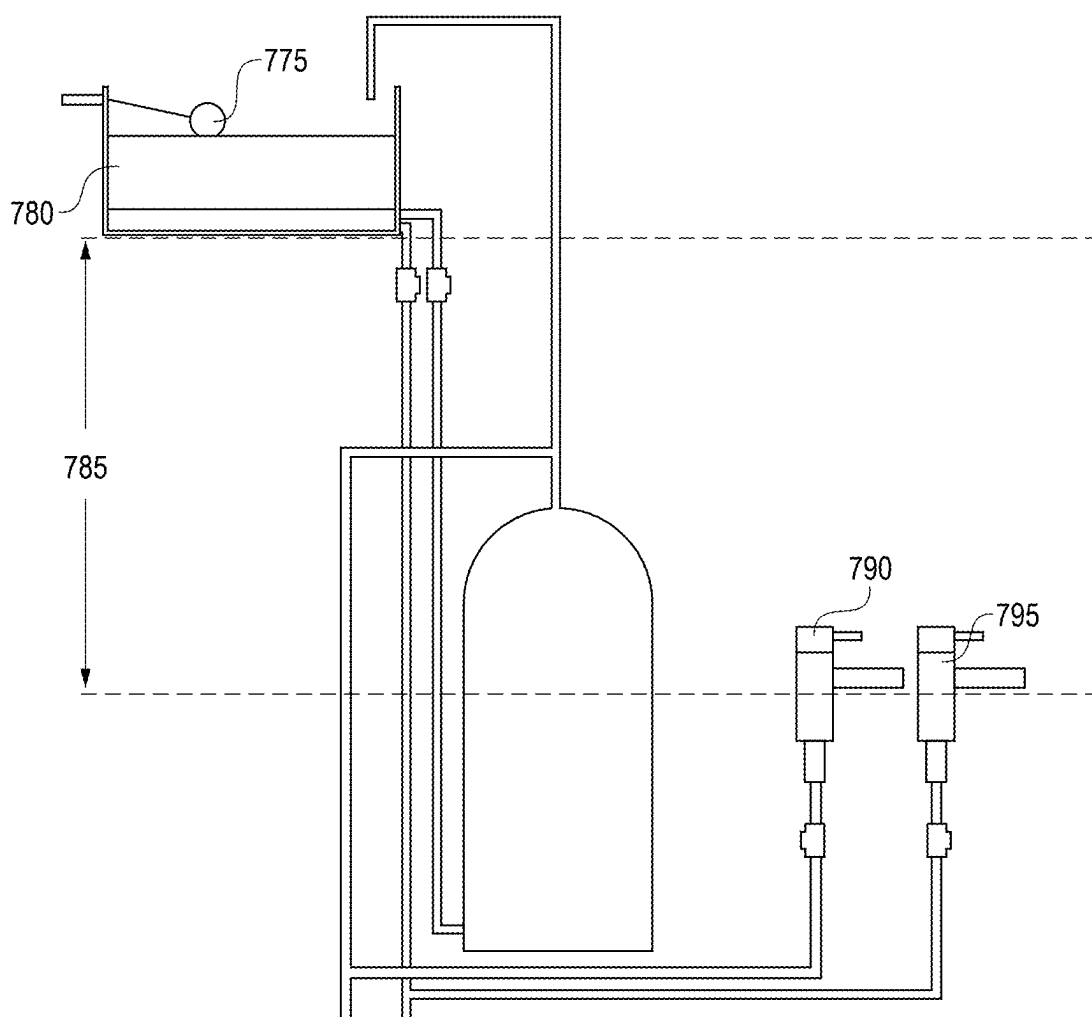


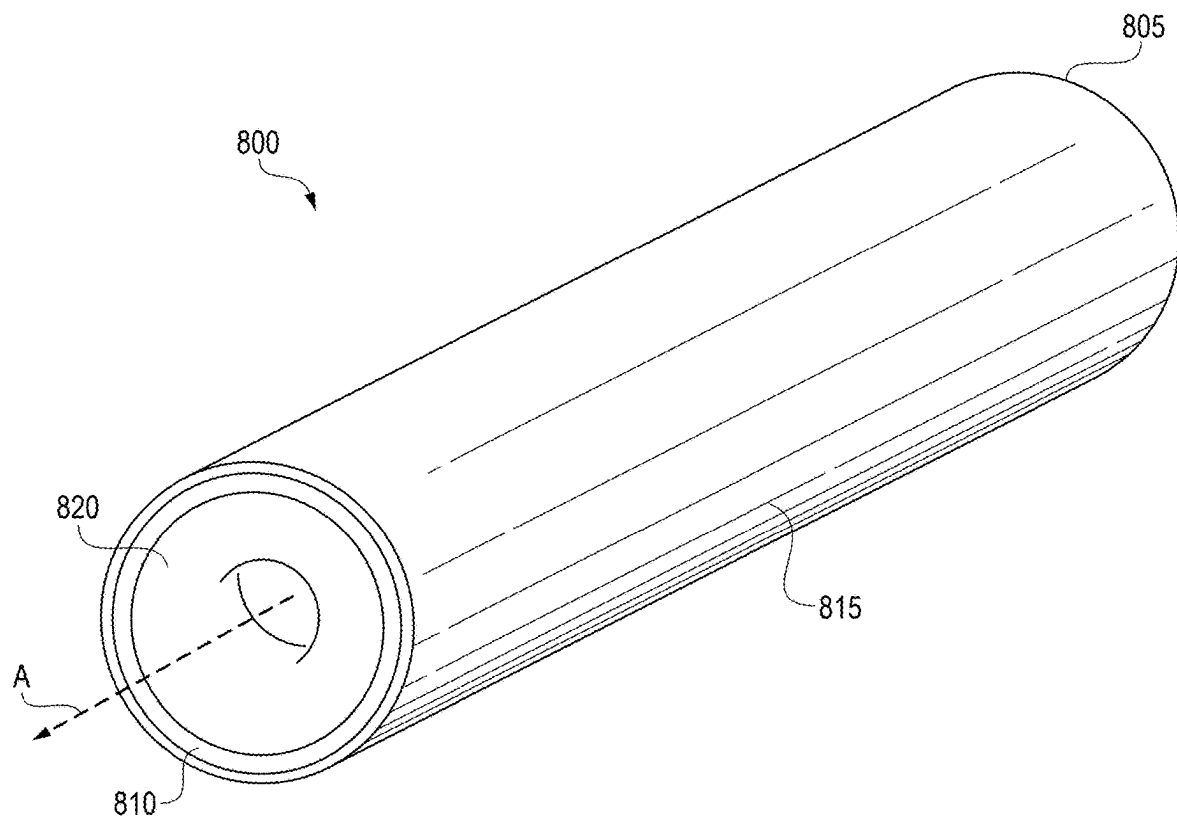
FIG. 26

FIG. 27

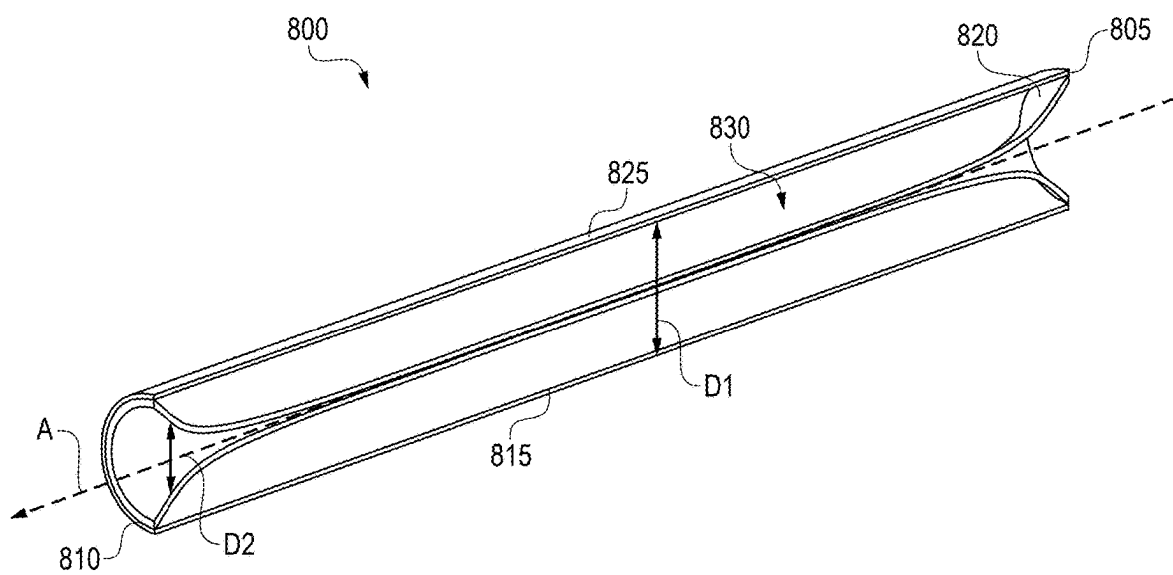
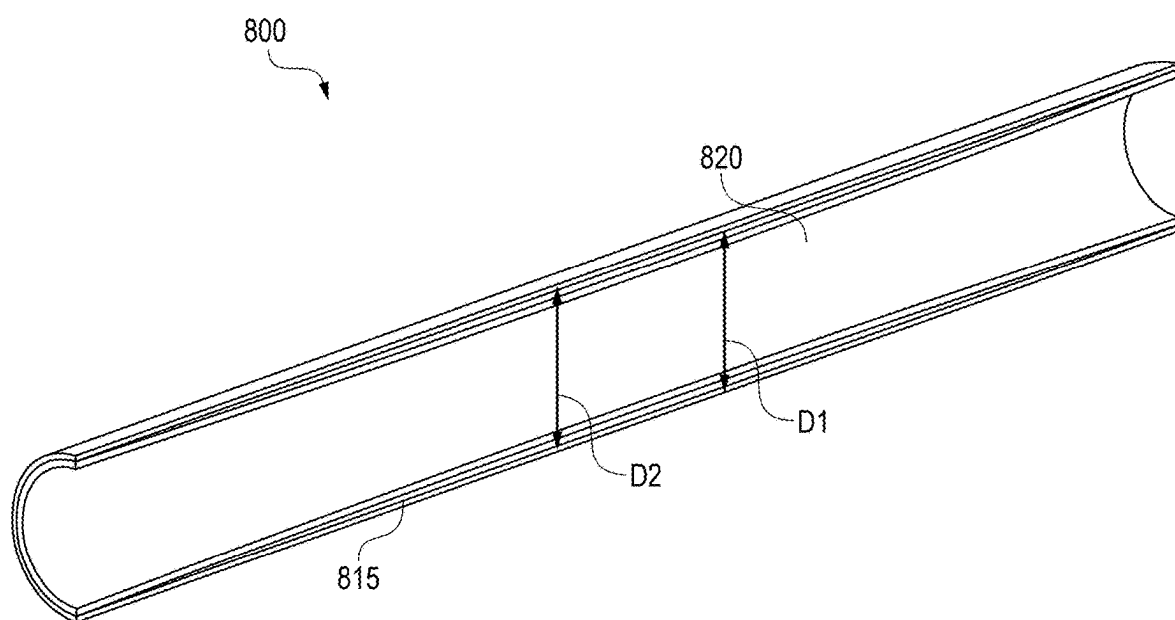


FIG. 28



WAIT WATER COLLECTION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/559,263, filed Feb. 29, 2024, entitled, “WAIT-WATER COLLECTION SYSTEM,” and to U.S. Provisional Patent Application No. 63/559,288, filed Feb. 29, 2024, entitled, “SYSTEM AND METHODS FOR MANAGEMENT OF WATER,” the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] At times, consumers may turn on a faucet (e.g., hot water valve) and wait for the water to reach a certain temperature before using the water. For example, consumers may wait to enter a shower until the water has reached a desired temperature. Water that has flowed through the system while the user waits for the temperature to warm up may enter the drain and septic system without use. The amount of this water, which may be referred to as “wait water,” may be considerable. It may be desirable for an automated system to collect the wait water for reuse in another device (e.g., toilet, cold water tap, or irrigation).

[0003] Further, stagnant water tends to occupy pipes that are not in use or are between uses. Generally, stagnant water is discarded when the pipe is next used. This may result in wasted water, higher utility payments, or other undesirable outcomes. Therefore, there is a need for a pipe that limits or eliminates the presence of stagnant water when the pipe is not in use.

BRIEF DESCRIPTION OF DRAWINGS

[0004] FIG. 1 is a schematic view of an example wait water collection system.

[0005] FIG. 2 is a perspective view of an example expansion vessel.

[0006] FIG. 3 is a perspective view of an example expansion vessel.

[0007] FIG. 4 is a front elevation view of an example drain gate.

[0008] FIG. 5 is a cross-sectional view of an example drain gate in a first stage of operation.

[0009] FIG. 6 is a cross-sectional view of an example drain gate in a second stage of operation.

[0010] FIG. 7 is a perspective view of an example venturi.

[0011] FIG. 8 is a cross-sectional view of an example venturi.

[0012] FIG. 9 is a schematic view of an example wait water collection system.

[0013] FIG. 10 is a schematic view of an example wait water collection system.

[0014] FIG. 11 is a perspective view of an example thermostatic diverter.

[0015] FIG. 12 is a side cross-sectional of an example thermostatic diverter.

[0016] FIG. 13 is a schematic view of an example wait water collection system.

[0017] FIG. 14 is a schematic view of an example wait water collection system.

[0018] FIG. 15 is a front elevation view of an example toilet with an example storage tank.

[0019] FIG. 16 is a schematic view of an example wait water collection system.

[0020] FIG. 17 is a schematic view of an example wait water collection system.

[0021] FIG. 18 is a schematic view of an example wait water collection system.

[0022] FIG. 19 is a schematic view of an example wait water collection system.

[0023] FIG. 20 is a schematic view of an example wait water collection system.

[0024] FIG. 21 is a schematic view of an example wait water collection system.

[0025] FIG. 22 is a schematic view of an example wait water collection system.

[0026] FIG. 23 is a schematic view of an example wait water collection system.

[0027] FIG. 24 is a schematic view of an example wait water collection system.

[0028] FIG. 25 is a schematic view of an example wait water collection system.

[0029] FIG. 26 is a perspective view of an example pipe with a flexible membrane.

[0030] FIG. 27 is a perspective cross-sectional view of an example pipe in a first stage of operation.

[0031] FIG. 28 is a perspective cross-sectional view of an example pipe in a second stage of operation.

[0032] While the disclosure is susceptible to various modifications and alternative forms, a specific embodiment thereof is shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

DETAILED DESCRIPTION

[0033] The invention will now be described with reference to the drawing figures, in which like reference numerals refer to like parts throughout. For purposes of clarity in illustrating the characteristics of the present invention, proportional relationships of the elements have not necessarily been maintained in the drawing figures.

[0034] As stated above, it may be desirable to have a wait water collection system to reduce water usage and water waste. A wait water collection system 1 may be a “one-box unit,” which may be installed to aid in preventing the waste of wait water. The wait water collection system 1 may include several components, as discussed herein, which may operate to aid in preventing the waste of wait water. The wait water collection system 1 may operate without additional electrical power, and the wait water collection system 1 may be operated using water pressure alone.

[0035] Hot water may enter a wait water collection system 1 via a hot water input 5. The hot water may travel via a first water line 10 to a thermal diverter 15, which may include a valve that adjusts based on a temperature. If the water meets or exceeds a threshold temperature, the thermal diverter 15 may cause the water to flow to a mixer valve 20 (which may have a hot water inlet 26 and a cold water inlet 30) via a second water line 35. The mixer valve 20 may be any mixer valve now known or hereafter developed which may be used to adjust a temperature of water. A user may select a

temperature setting, which may cause the wait water collection system 1 (and components therein) to adjust to output water at the selected temperature. The water mixed via the mixer valve 20 may exit the wait water collection system 1 via an outlet 40 (e.g., a showerhead, handset, faucet, etc.).

[0036] If the water is below a threshold temperature (which may be predetermined), the valve within the thermal diverter 15 may cause the water to flow through a third water line 45 into an expansion vessel 50 via an expansion vessel port 55. The expansion vessel 50 may be any suitable capacity. As one example, the expansion vessel 50 may have a capacity of 8 liters or 2.11 gallons; as another example, the expansion vessel 50 may have a capacity of 6 to 16 liters. The expansion vessel 50 may include an air inlet 60 which may be used to adjust the air pressure within the expansion vessel 50. The air pressure may be adjusted, for example, depending on the water pressure being delivered to the system. A user may access the air inlet 60 in order to alter the air pressure in the expansion vessel 50. Even when the expansion vessel 50 is not easily accessible by a user (e.g., behind a wall), the air inlet 60 may be attached to an air filler tube, which the user may access to alter the air pressure in the expansion vessel 50. The air filler tube may include a standard tyre valve, which may be connected to a standard pump and air pressure gauge to change the air pressure.

[0037] Cold water may enter the wait water collection system 1 via a cold water input 65. The cold water may flow through a fourth water line 70 and may pass through a venturi 75. The venturi 75 may be used to draw water out from the expansion vessel 50. The water from the expansion vessel 50 may exit the expansion vessel 50 via the expansion vessel port 55 and may flow to the venturi 75 via a fifth water line 80. The water from the expansion vessel 50 may mix with the water from the cold water input 65 and may be directed into the mixer valve 20 via a sixth water line 85.

[0038] A pressure sensor 90 may be positioned on the second water line 35. The pressure sensor 90 may determine a pressure of the water passing through the second water line 35, and the pressure detected by the pressure sensor 90 may be sent to each of an inlet gate valve 95 and a drain gate valve 100. The pressure may be sent via a pilot tube 105 connecting the pressure sensor 90 to each of the inlet gate valve 95 and the drain gate valve 100. When the water pressure is low, the inlet gate valve 95 may be opened and the drain gate valve 100 may be closed. When the water pressure is high, the inlet gate valve 95 may be closed and the drain gate valve 100 may be opened. When the inlet gate valve 95 is opened, hot water may flow in via the hot water input 5 to the thermal diverter 15. When the drain gate valve 100 is opened, water may flow out of the expansion vessel 50 through the outlet 40 via a seventh water line 110.

[0039] As a first example, when the wait water collection system 1 is in a “static” state (i.e., a previous user has switched off the mixer valve 20 when hot water was flowing), the pressure sensor 90 may detect a high pressure. As such, the inlet gate valve 95 may close, and the drain gate valve 100 may open such that water from the expansion vessel 50 flows out via the outlet 40. As a second example, when a user switches on the shower with a hot temperature selected, the pressure sensor 90 may detect a low pressure. As such, the inlet gate valve 95 may open, and the drain gate valve 100 may close such that water from the expansion vessel 50 flows into the venturi 75 and mixes with the cold

water. As a third example, when cold water flows into the expansion vessel 50, the pressure sensor 90 may detect a low pressure. As such, the inlet gate valve 95 may open, and the drain gate valve 100 may close such that water from the expansion vessel 50 flows into the venturi 75 and mixes with the cold water.

[0040] As a fourth example, when the hot leg arrives, the pressure sensor 90 may detect a low (but rising) pressure. As such, the inlet gate valve 95 may open, and the drain gate valve 100 may close such that water from the expansion vessel 50 flows into the venturi 75 and mixes with the cold water. As a fifth example, when the hot leg is flowing, the pressure sensor 90 may detect a medium (but not high) pressure. As such, the inlet gate valve 95 may open, and the drain gate valve 100 may close such that water from the expansion vessel 50 flows into the venturi 75 and mixes with the cold water. As a sixth example, when a user switches off the unit, the pressure sensor 90 may detect a high pressure. As such, the inlet gate valve 95 may close, and the drain gate valve 100 may open such that water from the expansion vessel 50 flows out via the outlet 40. At the end of the showering period, the expansion vessel 50 may be emptied. The foregoing operational examples are provided for exemplary purposes only and should not be construed as limited.

[0041] The pressure sensor 90 connectivity to the inlet gate valve 95 and the drain gate valve 100 may be useful in compensating for failures of the wait water collection system 1. For example, if a user switches off the mixer valve 20 before the hot leg arrives, the pressure sensor 90 may detect a low pressure, and as such, the drain gate valve 100 may be closed. Water may continue to flow into the expansion vessel 50 even though the mixer valve has been switched off. Once the hot leg arrives, the thermal diverter 15 may send the hot water to the mixer valve 20. The pressure sensor 90 may detect a high pressure, and as such, the drain gate valve 100 is opened. The expansion vessel 50 may then empty, and although some water may be wasted in the process, the wait water collection system 1 may be ready for a next user.

[0042] In the circumstance where the hot leg does not arrive (e.g., due to a fault of the water heater), the thermal diverter 15 may send ambient water to the expansion vessel 50. Once the expansion vessel 50 is filled, the wait water collection system 1 may lock. The wait water collection system 1 may be primed for a new user once the water heater fault is fixed. In order to be primed for a new user, a user may switch the mixer valve 20 “on” with a set temperature of cold, such that the water from the expansion vessel 50 is drained through the outlet 40 via the venturi 75 and mixer valve 20.

[0043] When in such a configuration, the expansion vessel 50 may be oriented vertically (as illustrated in FIG. 2) or horizontally (as illustrated in FIG. 3). In either orientation, a base 115 of the expansion vessel 50 may be slanted, such that water in the expansion vessel 50 drains toward the expansion vessel port 55. The expansion vessel port 55 may be positioned at an edge 120 of the expansion vessel 50 where the base 115 slants downward. Such a configuration may aid in ensuring that water does not collect in the expansion vessel 50 and that water is successfully drained out of the expansion vessel 50. It should be understood that any other suitable orientation or positioning may be achieved and that FIGS. 2 and 3 should not be construed as limiting.

[0044] Turning now to FIG. 4, the drain gate valve 100 as discussed herein may adjust based on the detected pressure by the pressure sensor 90. The drain gate valve 100 may be positioned on the seventh water line 110 near or adjacent to the second water line 35. Turning to FIG. 5, the drain gate valve 100 may include a rocker plunger 125, which may be operated based on the pressure of water flowing in the second water line 35. When high pressure is detected, a diaphragm 130 within the drain gate valve 100 may press in a downward direction on a first end 135 of the rocker plunger 125. This may cause a second end 140 of the rocker plunger 125 to lift, which may allow the drain gate valve 100 to open, causing the water in the expansion vessel 50 to drain.

[0045] As illustrated in FIG. 6, when low pressure is detected, the diaphragm 130 may remain in place, causing the second end 140 of the rocker plunger 125 to drop. This may close the drain gate valve 100, causing the water from the expansion vessel 50 to flow through the venturi 75 and mix with the cold water.

[0046] The venturi 75 as discussed herein may be used to draw water out of the expansion vessel 50. The venturi 75 may be any venturi now known or hereafter developed, and as one example, the venturi 75 may be the venturi as disclosed in U.S. patent application Ser. No. 18/947,734, filed Nov. 14, 2024, entitled, "POD VENTURI," the entire contents of which are hereby incorporated by reference. The venturi 75 may include multiple venturis for installation, as illustrated in FIG. 7. As one example, the venturi 75 may include a high pressure venturi 145 and a low pressure venturi 150, one of which may be installed for use.

[0047] When the wait water collection system 1 is being installed, an installer may determine whether the high pressure venturi 145 or the low pressure venturi 150 is suitable for the situation. If the installer determines that the water flowing in the system is at a high pressure, for example, the high pressure venturi 145 may be installed inline with the applicable water line. The venturi 75 may include an arrow 155 indicating which direction water should flow when the venturi 75 is installed, which may be useful for installation purposes. The venturi 75 may include an expansion vessel input 160 which may input water from the expansion vessel 50 into the venturi 75. When installed, the expansion vessel input 160 may be connected to the fifth water line 80. Each end of the venturi 75 may include surfaces where o-ring seals may be installed to aid in ensuring a tight seal with the water lines.

[0048] Turning to FIG. 8, the venturi 75 may include a primary water input 165 which may feed water from the cold water input 65 via the fourth water line 70. The venturi 75 may also include the expansion vessel input 170 which may feed water from the expansion vessel 50. The venturi 75 may include a water outlet 40 where the water mixed from the cold water input 65 and expansion vessel 50 may exit the venturi 75. When installed, such output may flow out of the venturi 75 into the sixth water line 85. The venturi 75 may include a larger circumference at each end 175 of the venturi 75 and a smaller circumference at a center point 180 of the venturi 75. The difference in the circumference may allow the water to reach a higher velocity while traveling through the venturi 75. The low pressure venturi 150 may include a larger center circumference than the high pressure venturi 145 with a smaller center circumference.

[0049] Turning to FIG. 9, the wait water collection system 1 may include similar components as those described above. However, the expansion vessel 50 may include a separate vessel input 185 and vessel output 190. In such a scenario, water may flow from the thermal diverter 15 into the expansion vessel 50 via the vessel input 185, and water may flow out of the expansion vessel 50 into the venturi 75 via the vessel output 190.

[0050] Turning now to FIG. 10, the wait water collection system 1 may include additional pressure sensors and/or thermal diverters, which may assist with overcoming certain failures. If the wait water collection system 1 is in normal operation, when a user switches on the mixer valve 20, a pressure of the second water line 35 may reduce. The pressure sensor 90 may sense such a reduction in pressure. As a result, the inlet gate valve 95 may open, which allows tepid water from the hot water input 5 to flow to the thermal diverter 15 and into the expansion vessel 50. Additionally, the drain gate valve 100 may close, which may cause the water collected in the expansion vessel 50 to flow to the venturi 75 if there is a cold water element of the user's temperature setting.

[0051] When the shower is switched off before the thermal diverter 15 has received water meeting or exceeding the temperature threshold, such water may travel to the expansion vessel 50. In such a situation, pressure in the second water line 35 may or may not be able to be re-pressurized. This may cause the inlet gate valve 95 to remain open, allowing water to continue flowing into the expansion vessel 50, even though the user has switched off the mixer valve 20 and left the shower. This may cause water to fill the expansion vessel 50. Because the drain gate valve 100 may be closed due to the low pressure in the second water line 35, the wait water collection system 1 could eventually fill and "lock up." In this condition, if the mixer valve 20 is switched "on," no more water can enter the expansion vessel 50 due to being full, and the wait water collection system 1 may not function as described above. To resume the process described above, the mixer valve 20 may be turned "on" and a cold temperature may be selected, which may cause the expansion vessel 50 to empty via the venturi 75. Then, a user may select a warmer temperature, which allows hot water to reach the thermal diverter 15 and re-pressurize the second water line 35. However, this process may be time consuming and lead to wasted water.

[0052] The wait water collection system 1 may include a pressure sensor 191 on the fourth water line 70. The pressure sensor 191 may detect a pressure from the cold water input 65. The pressure may be sent via a pilot tube 192 to a tepid gate valve 193 on the third water line 45. The detected pressure may cause the tepid gate valve 193 to open or close. Pressure may be higher when the shower is not flowing, while pressure may be lower when the shower is flowing. Such lower pressures may be as low as 0.1 Bar (1.45 psi), which may be detectable by the pressure sensor 191. When the pressure is higher, the tepid gate valve 193 may close, preventing water from entering the expansion vessel 50. When the pressure is lower, the tepid gate valve 193 may open, causing water to flow into the expansion vessel 50. This may aid in ensuring that water stops flowing to the expansion vessel 50 when the mixer valve 20 is switched "off."

[0053] A thermostatic diverter 195, as illustrated in FIG. 11, may be used to divert water to and from different outlets

based on temperature of media entering the diverter **195**. The thermostatic diverter **195** may be used in place of the thermal diverter **15** as discussed above with reference to FIGS. **1-10**. The thermostatic diverter **195** may include a supply feed **200**. The supply feed **200** may be a supply of any media (e.g., hot water). The water from the supply feed **200** may enter the thermostatic diverter **195** and may be diverted to a first outlet **205** and/or a second outlet **210**. The thermostatic diverter **195** may direct the media to only one of the first outlet **205** or the second outlet **210**, or to both of the first outlet **205** and the second outlet **210** simultaneously. The first outlet **205** may be coupled or connected to a storage tank. As one example, the storage tank may store cold water. The second outlet **210** may be coupled or connected to a faucet (e.g., shower, sink faucet, etc.). The thermostatic diverter **195** may cause water that reaches a certain or desired temperature to exit the diverter **195** via the second outlet **210**, while water that has not reached such certain temperature is directed to exit the diverter **195** via the first outlet **205**.

[0054] Turning to FIG. **12**, a diaphragm **215** may control flow into and through the thermostatic diverter **195**. As such, when the diaphragm **215** is in a closed position due to a valve (e.g., shower valve, faucet valve, etc.) connected to the diverter **195** being closed, no fluid (e.g., water) may flow through the thermostatic diverter **195**. As illustrated in FIG. **12**, the diaphragm **215** is in a closed position. The thermostatic diverter **195** may be activated when flow is initiated by opening the valve connected to the thermostatic diverter **195**. When flow is initiated, pressure at a first end **220** of a check valve **225** may drop, which may cause a diaphragm **215** to adjust to an open position due to a pressure imbalance. When in a closed position due to the shower valve, faucet valve, etc.

[0055] being closed, the check valve **225** may trap pressure between the check valve **225** and the diaphragm **215**, which may close the diaphragm **215**. When the diaphragm **215** is closed, water may be prevented from flowing into or out of the thermostatic diverter **195**.

[0056] When the diverter **195** is activated via the diaphragm **215** opening, water, which may flow into the diverter **195** via the supply feed **200**, may pass over a thermostat **230**. When water has not reached a certain temperature, the thermostat **230** may bias a shuttle **235** to direct water to the first outlet **205**, which may feed, for example, into a storage tank. When water has reached a certain temperature, the thermostat **230** may bias a shuttle **235** to direct water to the second outlet **210**. The thermostat **230** may extend to bias the shuttle **235** in such a direction. As one example, the temperature at which the water should be diverted to the second outlet **210** is pre-set or pre-determined. As another example, an adjustment system **240** may be used to select a temperature at which the water should be diverted from the first outlet **205** to the second outlet **210**. A user may input a desired temperature into the adjustment system **240** (e.g., through a user interface), which may be coupled to control translation of the thermostat **230**.

[0057] Turning now to FIG. **13**, water may enter a bathroom segment **245** of a wait water collection system via an input line **250**. The entire wait water collection system may include a combination of the components described herein. The water may be cold water from a municipal water supply. The input line **250** may feed water to a shower **255**, storage

tank **260**, and/or any other desirable outlet. The input line **250** may feed water through a heater **265** to be heated. While being heated or while stored in a diverter input line **270**, the water may not necessarily be a desired temperature. For example, after a shower is shut off, water may sit within the diverter input line **270** and cool off. To determine whether the water has reached said desired temperature, the water may be fed into a thermostatic diverter **275** via the diverter input line **270**. The thermostatic diverter **275** may be similar to or the same as the thermostatic diverter **195** as discussed with reference to FIGS. **11** and **12**.

[0058] If the water has reached the desired temperature, the water may travel to the shower **255** via a hot water line **280**. If the water has not reached the desired temperature, the water may be stored in the storage tank **260** for later or other use. The water in the storage tank **260** may provide water to a toilet **285** or to any other desirable outlet (e.g., a cold water faucet). After used in the shower **255**, toilet **285**, or other outlet, the water may flow to a septic system via septic lines **290**.

[0059] Instead of to the storage tank **260** before being directed to the toilet **285**, as illustrated in FIG. **14**, water may be directly fed into a toilet **295** from the thermostatic diverter **275**. In such configuration, the toilet **295** may include a toilet storage tank **300** and a flush tank **305**, as illustrated in FIG. **15**. The toilet storage tank **300** may sit on top of the flush tank **305**, may be placed in a wall, or may take any other suitable configuration now known or hereafter developed. The toilet **295** may include a diversion valve which may be used to adjust whether water from the toilet storage tank **300** or the flush tank **305** is used in a toilet bowl **310**.

[0060] As illustrated in FIG. **16**, hot water may be stored in a highly insulated hot water storage tank **315**. Water may be directed from a supply input **320** to a heater **325** in order to be heated. The water may be fed to the storage tank **315** via a storage input line **330**. Even if the water from the heater **325** is not sufficiently heated, the water may mix with the hot water in the storage tank **315** in order to be adequately heated to a desired temperature. Water from the input **320** and from the storage tank **315** may be fed to a shower **335**. After use in the shower **335**, water may flow to a septic via a septic line **340**.

[0061] Turning now to FIG. **17**, hot water may enter a wait water collection system **345** via a hot water input line **350**, and cold water may enter the wait water collection system **345** via a cold water input line **355**. The cold water may be directed directly to a shower mixer valve **360**. The hot water may be fed into a thermostatic diverter **365**, which may be similar to or the same as the thermostatic diverter **395** of FIGS. **11** and **12**. When a valve associated with a shower outlet **370** is opened, pressure (via a pressure feedback loop **375**) may cause a check valve **380** of the diverter **365** to adjust. A diverter switch **385** may turn “on” and cause water to flow into the diverter **365**.

[0062] If water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir **390**. From the reservoir **390**, water may be directed to a check valve **395** and into a venturi **400**. The venturi **400** may be used to introduce a slow flow rate throughout the use of the shower. If the water has reached the desired temperature, then the water may be fed into the shower mixer valve **360**. The shower mixer valve **360** may mix the cold water and the hot water and may direct the mixed water to the venturi **400**.

From the venturi 400, water may exit the wait water collection system 345 via the shower outlet 370. Such a configuration may be used as a mechanical mixer.

[0063] As illustrated in FIG. 18, hot water and cold water may be directed into the shower mixer valve 405 of a wait water collection system 410 via a hot water input line 415 and a cold water input line 420, respectively. The mixed water may then be directed to a thermostatic diverter 425, which may be similar to or the same as the thermostatic diverter 195 of FIGS. 11 and 12. The diverter 425 may determine whether the mixed water has reached a desired temperature.

[0064] If the mixed water has not reached the desired temperature, then the insufficiently heated water may be directed to a reservoir 430. The water may then be directed to a check valve 435 and to a venturi 440. The venturi 440 may be used to introduce a slow flow rate throughout the use of the shower. If the mixed water has reached the desired temperature, then the water may be directed from the mixer valve 425 to the venturi 440. The water from the reservoir 430 and/or the thermostatic diverter 425 may be fed to a shower outlet 445 via the venturi 440. Such a configuration may be used as a mechanical mixer.

[0065] Turning to FIG. 19, hot water may be inputted into a wait water collection system 450 via a hot water input line 455. The hot water may be fed into a thermostatic diverter 460, which may be similar to or the same as the thermostatic diverter 195 of FIGS. 11 and 12. When a valve associated with a shower outlet 465 is opened, pressure (via a pressure feedback loop 470) may cause a check valve 475 of the diverter 460 to adjust. A diverter switch 480 may turn “on” and cause water to flow into the diverter 460. If water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir 485. From the reservoir 485, water may be directed to a pump 490 and then to a check valve 495.

[0066] If the water has reached the desired temperature, then the water may be fed into a shower mixer valve 500 from the diverter 460. The shower mixer valve 400 may mix the hot water from the thermostatic diverter 460 and cold water, which may be fed into the system 450 via a cold water input line 505 passing through a check valve 510. Water may exit the wait water collection system 450 via the shower outlet 465. Such a configuration may be used as a mechanical mixer with a preferred reintroduction point using a pump.

[0067] As illustrated in FIG. 20, hot water and cold water may be inputted into a wait water collection system 515 via a hot water input line 520 and a cold water input line 525, respectively. The system 515 may include a hot water pressure limiter 530 along the hot water input line 520 prior to a diverter switch 535. The system 515 may include a cold water pressure limiter 540 along the cold water input line 525 between a check valve 545 and a venturi 550. The hot water pressure limiter 530 and cold water pressure limiter 540 may be any pressure limiter known in the art, including a Neoperl® pressure limiter. The hot water pressure limiter 530 and cold water pressure limiter 540 may adjust the pressure of the hot water in the hot water input line 520 or cold water in the cold water input line 525, respectively, to a pressure set point, which may be selected by a user. The venturi 550 may be used to introduce a slow flow rate throughout the use of the shower.

[0068] The hot water may be fed from the hot water pressure limiter 530 into a thermostatic diverter 555, which may be similar to or the same as the thermostatic diverter 195 of FIGS. 11 and 12. When a valve associated with a shower outlet 560 is opened, pressure (via a pressure feedback loop 565) may cause a check valve 570 of the diverter 555 to adjust. The diverter switch 535 may turn “on” and cause water to flow into the diverter 555. If the water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir 575. From the reservoir 575, water may be directed to a check valve 580, to the venturi 550, and to a shower mixer valve 585. If the water has reached the desired temperature, then the water may be fed into the shower mixer valve 585 from the check valve 570. The shower mixer valve 585 may mix the hot water and cold water. Water may exit the wait water collection system 515 via the shower outlet 560. Such a configuration may be used as a mechanical mixer with a preferred reintroduction point using a venturi.

[0069] As illustrated in FIG. 21, a wait water collection system 590 may utilize a digital mixer scheme. Hot water may be fed into the system 590 from a hot water input line 595. The hot water may be fed through a temperature sensor 600 and into a digital mixer valve dual outlet 605, and cold water may be fed directly into the digital mixer valve 605. The temperature sensor 600 may be coupled to the digital mixer valve 605 to determine whether the water passing into the digital mixer valve 605 has reached a certain temperature. If so, then the water may flow through a venturi 610 and a temperature sensor 615 to exit via a shower outlet 620. The temperature sensor 615 may provide feedback to the digital mixer valve 605 regarding temperature of the water passing out of the digital mixer valve 605. If not, then the water may flow into a reservoir 625, which may then flow through a check valve 630 into the venturi 610 to exit via the shower outlet 620.

[0070] Turning now to FIG. 22, a wait water collection system 635 may utilize a digital mixer scheme with a preferred reintroduction point. Cold water may be fed through a check valve 640 and into a digital mixer valve 645, and hot water may be fed through a temperature sensor 650 and into the digital mixer valve 645. The temperature sensor 650 may be coupled to the digital mixer valve 645 to determine whether the water passing through the digital mixer valve 645 has reached a certain temperature. If so, then the water may flow out via shower outlets 655. If not, then the water may flow into a reservoir 660 for storage. If needed, the water may flow from the reservoir 660 through a check valve 665 and may be reintroduced to the digital mixer valve 645 via a pump 670.

[0071] As illustrated in FIG. 23, a wait water collection system 675 may include a hot water pressure limiter 680 along a hot water input line 685. The hot water input line 685 may input hot water into the system 675, and the hot water may pass through a temperature sensor 690 after passing through the hot water pressure limiter 680. Cold water may be introduced into the system 675 via a cold water input line 695. The system 675 may include a cold water pressure limiter 700 along the cold water input line 695 between a check valve 705 and a venturi 710. The hot water pressure limiter 680 and cold water pressure limiter 700 may be any pressure limiter known in the art, including a Neoperl® pressure limiter. The hot water pressure limiter 680 and cold water pressure limiter 700 may adjust the pressure of the hot

water in the hot water input line **685** or cold water in the cold water input line **695**, respectively, to a pressure set point, which may be selected by a user.

[0072] The cold water and the hot water may be fed into a digital mixer valve **715** from the venturi **710** and the temperature sensor **690**, respectively. The venturi **710** may be used to introduce a slow flow rate throughout the use of the shower. The temperature sensor **690** may be coupled to the digital mixer valve **715** to provide an indication as to whether the water passing through the digital mixer valve **715** has reached a certain temperature. If so, then the water may flow out via shower outlets **720**. If not, then the water may flow into a reservoir **725** for storage. If needed, the water may flow from the reservoir **725** through a check valve **730** and may be reintroduced to the digital mixer valve **715** via the venturi **710**. Such a configuration may be used as a digital mixer with a preferred reintroduction point using a venturi.

[0073] Turning now to FIG. 24, a water supply **735** may provide water via a cold water line **740** to a water heater **745**, a washer **750**, a plurality of sinks **755**, and/or a shower **760**. The water heater **745** may heat the water from the water supply **735** and may provide heated water via a hot water line **765** to the washer **750**, the plurality of sinks **755**, and/or the shower **760**. Water may be stored in accumulator tanks **770** associated with the sinks **755** and/or the shower **760**. The accumulator tanks **770** may be used to store the water until a desired temperature of the water is reached, and the stored water, which may not have reached the desired temperature, may be used as an input to the cold water line **740**.

[0074] As illustrated in FIG. 25, a float valve **775** may be used within a reservoir **780** to determine whether the reservoir **780** is full. If the float valve **775** indicates that the reservoir **780** is full, then water may be directed away from the reservoir **780**. The reservoir **780** may be positioned a certain distance **785** between both of a hot water outlet **790** and a cold water outlet **795**, and such distance **785** may be associated with the static head pressure.

[0075] Turning now to FIG. 26, a water management pipe **800** may provide water from a source to a user. The pipe **800** may be substantially cylindrical in shape and may include a first end **805** and a second end **810** opposing the first end **805**. The first end **805** may receive, be received by, or otherwise be coupled to an external source (not shown) of water or other fluid, which will collectively be referred to hereinafter as “water.” Alternatively, the first end **805** may be attached to separate piping, and that separate piping may be connected to an external water source, such that the pipe **800** is positioned downstream of the water source. The second end **810** may provide water to or toward a user. Alternatively, the second end **810** may receive, be received by, or otherwise be coupled to an appliance, attachment, or other mechanism (not shown) for providing water. For example, the second end **810** may be in fluid communication with a showerhead, faucet, or any other suitable means for supplying water to a user. As with the first end **805**, the second end **810** may be attached to separate piping, and that separate piping may be deliver water to a user or to another end use, such that pipe **800** is positioned upstream of the user or end use. Thus, water may enter the pipe **800** at the first end **805**, travel through the pipe **800** along a path that is substantially aligned with a central longitudinal axis A of the

pipe **800**, and exit the pipe **800** at the second end **810** where it may be provided to a user. The pipe **800** may include bends or curves.

[0076] The pipe **800** may include a shell **815** and a flexible membrane **820** positioned within the shell **815**. The shell **815** may be substantially cylindrical in shape, or another shape as desired. The shape and structure of the shell **815** may be fixed, whereas the shape and structure of the flexible membrane **820** may vary. In general, a cross section of the flexible membrane **820** taken in a direction substantially perpendicular to the axis A at any point along the pipe **800** may have a generally circular profile, or any other fixed profile as desired. However, the shape and structure of the flexible membrane **820** may vary depending on the flow of water through the pipe **800**. Additionally, the shell **815** and the flexible membrane **820** may be provided as coaxial structures. For example, the shell **815** and flexible membrane **820** may each be formed around and extend along the same central axis A.

[0077] The flexible membrane **820** may be coupled to the shell **815** at the first end **805** and the second end **810** of the pipe **800**, or anywhere in between. Thus, while the shape and structure of the flexible membrane **820** between the first and second ends **805**, **810** may change during various stages of operation of the pipe **800**, the flexible membrane **820** may be securely positioned within the shell **815**. The flexible membrane **820** may be coupled to the shell **815** such that, when the pipe **800** is in use, water may flow through the space defined by the flexible membrane **820**. For example, when the pipe **800** is in use, water may be excluded from a space located between the flexible membrane **820** and the shell **815**. Instead, water may be restricted to flowing through the flexible membrane **820**.

[0078] Turning to FIGS. 27 and 28, the shape and structure of the flexible membrane **820** may differ between a first stage of operation and a second stage of operation of the pipe **800**. During the first stage of operation, illustrated in FIG. 27, water may not be flowing through the pipe **800**. The flexible membrane **820** may occupy an idle position during the first stage of operation. During the second stage of operation, illustrated in FIG. 28, water may be flowing through the pipe **800**. The flexible membrane **820** may occupy an active position during the second stage of operation.

[0079] The flexible membrane **820** may include or be formed from a phase change material (PCM). The PCM may be provided in the form of a salt hydrate, paraffin wax, non-paraffin organic, metal alloy, or any other suitable material. The PCM may facilitate the flexible membrane **820** transitioning from the idle position to the active position. For example, the PCM may cause or allow the shape and structure of the flexible membrane **820** to change in response to the flow of water through the pipe **800**.

[0080] As illustrated in FIG. 27, during the first stage of operation, the flexible membrane **820** may occupy an idle position. In the idle position, the flexible membrane **820** may reduce the effective internal diameter of the pipe **800**. For example, the shell **815** may have an internal diameter D1. The internal diameter D1 of the shell **815** may be substantially constant throughout the pipe **800**. Alternatively, the internal diameter D1 may vary. The flexible membrane **820** may reduce the effective internal diameter of the pipe **800** to an amount less than the internal diameter D1 of the shell **815**. For example, an internal diameter D2 of the flexible

membrane 820 may be less than the internal diameter D1 of the shell 815. Additionally, the internal diameter D2 may vary at different points along the pipe 800. At the first and second ends 805, 810 where the flexible membrane 820 is coupled to the shell 815, the internal diameter D2 of the flexible membrane 820 may be substantially equal to or minimally less than the internal diameter D1 of the shell 815. Thus, at the first and second ends 805, 810, the effective internal diameter of the pipe 800 may be reduced by a minimal amount or not at all.

[0081] However, as the flexible membrane extends away from the first and second ends 805, 810 and toward a midpoint 825 of the pipe 800, the internal diameter D2 of the flexible membrane 820 may decrease. As the internal diameter D2 decreases, the effective internal diameter of the pipe 800 may be reduced and the flexible membrane may at least partially restrict the flow of water through the pipe 800. In the example of FIG. 27, as the flexible membrane 820 approaches the midpoint 825, the internal diameter D2 may decrease to essentially zero or close to zero. Thus, as the flexible membrane 820 approaches the midpoint 825, the effective internal diameter of the pipe 800 may be reduced to essentially zero or close to zero, thereby restricting the rate of flow of water through the pipe 800 to essentially zero or close to zero.

[0082] In addition to the example of FIG. 27, the internal diameter D2 of the flexible membrane 820 at the midpoint 825 may have any other value less than the internal diameter D1 of the shell 815. Alternatively, the internal diameter D2 of the flexible membrane 820 at the midpoint 825 may have any other value less than the internal diameter D2 of the flexible membrane 820 at the first and second ends 805, 810.

[0083] The flexible membrane 820 may be imparted with an inherent structural bias toward the idle position illustrated in FIG. 27. For example, the construction of the flexible membrane 820 may apply an inward pressure that biases the flexible membrane 820 away from the shell 815 and toward the axis A. Thus, when no external force or stimulus is applied, the flexible membrane 820 may default to occupying the idle position. When the flexible membrane 820 is in the idle position, the pipe 800 may include dead space 830 positioned between the flexible membrane 820 and the shell 815. For example, as the internal diameter D2 of the flexible membrane 820 decreases, a volume of the dead space 830 may increase. Water may be excluded from the dead space 830 or may be restricted from entering the dead space 830. In this way, the flexible membrane 820 may prevent or restrict the accumulation of stagnant water within the pipe 800 when the pipe 800 is not in use.

[0084] However, the flexible membrane 820 may change shape in response to external forces or stimuli. For example, when water flows into the pipe 800 via the first end 805, the flowing water may apply an outward pressure to the flexible membrane 820. The outward pressure applied by the flowing water may directly or indirectly oppose the inherent structural bias of the flexible membrane 820 toward the idle position. Thus, pressure applied to the flexible membrane 820 by the flowing water may tend to cause the flexible membrane 820 to move from the idle position toward the active position. The pressure may increase the internal diameter D2 of the flexible membrane 820 and may permit water to flow more freely from the first end 805 to the second end 810.

[0085] Additionally, the PCM of the flexible membrane 820 may contribute to or facilitate the flexible membrane 820 moving from the idle position to the active position. The PCM may cause the flexible membrane 820 to change shape more readily in response to external stimuli. For example, heated water may enter the pipe 800 via the first end 805 and come into contact with the flexible membrane 820. The PCM of the flexible membrane 820 may absorb thermal energy from the heated water. This thermal energy may, for example, cause the flexible membrane 820 to become more malleable. In this way, the heated water may cause the flexible membrane 820 to be more susceptible to being pushed toward the shell 815 by the pressure of water flowing through the pipe 800.

[0086] If water ceases flowing through the pipe 800, the PCM of the flexible membrane 820 may release thermal energy, causing the flexible membrane 820 to become less malleable. Additionally, the cessation of water flowing through the pipe 800 may eliminate or reduce the outward pressure applied to the flexible membrane 820. As the flexible membrane 820 loses malleability, and without the outward pressure of water flowing through the pipe 800, the inherent structural bias of the flexible membrane 820 may cause the flexible membrane 820 to return to the idle position.

[0087] Turning to FIG. 28, the flexible membrane 820 may occupy the active position during the second stage of operation of the pipe 800. In the active position, the shape of the flexible membrane 820 may be substantially cylindrical and may mirror the shape of the shell 815. Thus, when the flexible membrane 820 is in the active position, the flexible membrane 820 may contact the shell 815 and the dead space 830 of FIG. 27 may be substantially or entirely eliminated. For example, in the active position, the internal diameter D2 of the flexible membrane 820 may be substantially constant throughout the pipe 800. Additionally, in the active position, the internal diameter D2 of the flexible membrane 820 may be substantially equal to or minimally less than the internal diameter D1 of the shell 815 throughout the pipe 800. In the second stage of operation of the pipe 800, the flexible membrane 820 may be maintained in the active position by the outward pressure applied by the flowing water, by the enhanced malleability or other characteristic of the flexible membrane 820 caused by the PCM, or by a combination thereof. Thus, in the second stage of operation, water may flow freely through the pipe 800.

[0088] The aforementioned storage tanks and reservoirs may be used to store cold water or hot water. As discussed above, water in a hot water input line may cool off while within the line after a shower is shut off, specifically if the climates surrounding the hot water input line are colder. Similarly, water in a cold water input line may warm up while within the line after a shower is shut off, specifically if the climates surrounding the cold water input line are warmer. It should be understood that the storage tanks and reservoirs, although discussed with reference to storing cold water, may store any temperature of water. Further, in order to aid in preventing bacterial growth mitigation in a water storage tank, certain cleansing of the water may occur prior to entering the tank. As one example, ultraviolet light may be used to cleanse the water.

[0089] As is evident from the foregoing description, certain aspects of the present invention are not limited by the particular details of the examples illustrated herein, and it is

therefore contemplated that other modifications, applications, variations, or equivalents thereof, will occur to those skilled in the art. Many such changes, modifications, variations, and other uses and applications of the present constructions will, however, become apparent to those skilled in the art after considering the specification and the accompanying drawings. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. All such changes, modifications, variations, and other uses and applications which do not depart from the spirit and scope of the present inventions are deemed to be covered by the inventions which are limited only by the claims which follow.

1. A wait water collection system comprising:
 - at least one water input coupled to a water line;
 - a thermal diverter positioned on the water line, wherein the thermal diverter adjusts based on whether a temperature of water from the at least one water input meets a temperature threshold;
 - an expansion vessel;
 - a mixer valve; and
 - an outlet;
 - wherein when the temperature of the water meets or exceeds the temperature threshold, the water is directed to the mixer valve; and
 - wherein when the temperature of the water falls below the temperature threshold, the water is directed to the expansion vessel.
2. The wait water collection system of claim 1, further comprising a venturi for drawing water out of the expansion vessel.
3. The wait water collection system of claim 1, further comprising a pressure sensor for determining a pressure within the water line.
4. The wait water collection system of claim 3, further comprising an inlet gate valve and a drain gate valve.
5. The wait water collection system of claim 4, wherein when the pressure sensor determines that the pressure is high, the inlet gate valve closes and the drain gate valve opens.
6. The wait water collection system of claim 4, wherein when the pressure sensor determines that the pressure is low, the inlet gate valve opens and the drain gate valve closes.
7. The wait water collection system of claim 4, wherein when the drain gate valve is open, water from the expansion vessel flows directly out of the wait water collection system via the outlet.
8. The wait water collection system of claim 4, wherein when the drain gate valve is closed, water from the expansion vessel flows to the mixer valve.
9. A wait water collection system comprising:
 - a first media input providing a first media to a first media line;
 - a second media input providing a second media to a second media line;
 - a thermal diverter positioned on the first media line, wherein the thermal diverter adjusts based on whether a temperature of the first media meets a temperature threshold;
 - a pressure sensor positioned on the first media line, wherein the pressure sensor determines a pressure within the first media line;

an expansion vessel, wherein when the temperature of the first media falls below the temperature threshold, the thermal diverter causes the first media to flow into the expansion vessel;

a venturi positioned on the second media line, wherein when the first media fills the expansion vessel, the venturi draws the first media out of the expansion vessel into the second media line;

a mixer valve coupled to each of the first media line and the second media line, wherein the first media and the second media mixes into a water output; and

an outlet for providing the water output to a user.

10. The wait water collection system of claim 9, further comprising an inlet gate valve and a drain gate valve.

11. The wait water collection system of claim 10, wherein when the pressure sensor determines that the pressure is high, the inlet gate valve closes and the drain gate valve opens.

12. The wait water collection system of claim 10, wherein when the pressure sensor determines that the pressure is low, the inlet gate valve opens and the drain gate valve closes.

13. The wait water collection system of claim 10, wherein when the drain gate valve is open, the first media from the expansion vessel flows directly out of the wait water collection system via the outlet.

14. The wait water collection system of claim 10, wherein when the drain gate valve is closed, the first media from the expansion vessel flows to the mixer valve via the second media line.

15. The wait water collection system of claim 9, wherein when the temperature of the first media meets or exceeds the temperature threshold, the first media is directed to the mixer valve.

16. The wait water collection system of claim 9, wherein the first media is hot water, and wherein the second media is cold water.

17. A wait water collection system comprising:

at least two water inputs coupled to at least two water lines;

a thermal diverter positioned on one of the at least two water lines, wherein the thermal diverter adjusts based on whether a temperature of water from the one of the at least two water inputs meets a temperature threshold; and

an expansion vessel;

wherein when the temperature of the water falls below the temperature threshold, the water is directed to the expansion vessel.

18. The wait water collection system of claim 17, wherein the expansion vessel has a capacity of 6 to 16 liters.

19. The wait water collection system of claim 17, wherein the expansion vessel includes an expansion vessel port for at least one of inputting and outputting water into the expansion vessel.

20. The wait water collection system of claim 17, wherein when a pressure in the wait water collection system is high, a drain gate valve is opened, and water from the expansion vessel flows out of the wait water collection system via an outlet.

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